



Mastercard SecureCode

Acquirer Implementation Guide

21 March 2017

Summary of Changes, 21 March 2017

This document reflects changes effective since the last publication of this manual..

Description of Change	Where to Look
Updated the Mastercard Secure Code And Identity Check Liability Shift Matrix	See the attached spreadsheet
Updated the chart title from "Liability Shift—Consumer Cards" to "Liability Shift—Cards and Commercial Cards"	Liability Shift Rule Summary
Removed chart "Liability Shift —Commercial Cards"	
Updated "Liability Shift" definition	Mastercard SecureCode Merchant Process and Liability Shift Matrix

Contents

Summary of Changes, 21 March 2017.....	2
Chapter 1: SecureCode Acquirer Implementation Overview.....	6
Mastercard and E-commerce.....	7
Maestro and E-commerce.....	7
Mastercard Debit.....	8
Mastercard SecureCode Program.....	8
Typical Payment Transaction Participants.....	10
Mastercard SecureCode Program Platform.....	11
What is UCAF and Its Structure?.....	11
UCAF Transport in Mastercard Authorization Messages.....	13
Sample DE 48 Contents.....	14
What is an AAV?.....	15
What is a Merchant Plug-In.....	16
Rules to Support UCAF Liability Shift.....	16
Liability Shift Rule Summary.....	17
UCAF Compliance Requirements.....	17
Authorization Requirements.....	17
Debit/Single Message System Requirements.....	20
Clearing Requirements.....	20
Enrollment Requirements.....	20
Maintenance Requirements.....	21
Participation Responsibilities for all Customers.....	21
Acquirer Responsibilities.....	22
Mastercard SecureCode Compliance and Functional Testing.....	23
Mastercard SecureCode Program Enrollment Procedure.....	23
Mastercard Support.....	24
Additional References.....	24
Chapter 2: 3-D Secure Solution.....	26
3-D Secure Solution Overview.....	27
Components.....	27
Issuer Domain.....	27
Acquirer Domain.....	29
Interoperability Domain.....	29
3-D Secure Solution Message Types.....	31
Card Range Request/Response.....	31
Verification Request/Response.....	31

Payer Authentication Request/Response.....	32
Payer Authentication Transaction Request/Response.....	32
Cardholder Enrollment.....	32
Cardholder Enrollment Process.....	32
Cardholder Authentication.....	33
Sample Cardholder Authentication Process.....	33
Chapter 3: Mastercard <i>SecureCode</i> Cardholders.....	37
Cardholder Overview.....	38
Cardholder Infrastructure.....	38
Cardholder Customization.....	38
Cardholder Operational.....	38
Accountholder Authentication Value Processing.....	39
Cardholder Global Infrastructure Testing Requirements.....	39
Chapter 4: Customer Interface Requirements.....	40
Purchase Authentication Page.....	41
Activation During Shopping Enrollment.....	41
Chapter 5: <i>SecureCode</i> Acquirers.....	43
Acquirer Overview.....	44
Acquirer Infrastructure.....	44
Establishment of <i>SecureCode</i> Operating Environment.....	44
Digital Signature Key Management.....	45
Authorization System Message Enhancements.....	45
Maestro Considerations.....	45
Acquirer Customization.....	47
Authentication Window Contents.....	47
Mastercard <i>SecureCode</i> Program Identifier Usage Guidelines.....	47
Acquirer Operational.....	48
Mastercard <i>SecureCode</i> Directory Server Maintenance.....	48
SSL and Digital Signing Certificates.....	48
Mastercard <i>SecureCode</i> 360 Webinar Series.....	48
Mastercard <i>SecureCode</i> Merchant Information.....	49
Mastercard <i>SecureCode</i> Merchant Process and Liability Shift Matrix.....	49
Chapter 6: Issuer Authentication Options.....	51
Issuer Authentication Options Background.....	52
Static Password.....	52
Random Static.....	53

Short Message Service (SMS).....	54
Mobile Application.....	55
Chip and Pin.....	56
Display Card.....	56
Other Authentication Options.....	57
Appendix A: Mastercard <i>SecureCode</i> Contact Information.....	59
Mastercard <i>SecureCode</i> Contact Information.....	60
Appendix B: Maestro Processing Considerations.....	62
Account in Good Standing.....	63
Appendix C: Mastercard Advance Registration Program Requirements.....	64
Maestro Advance Registration Program.....	65
MARF Merchant Use of Mastercard <i>SecureCode</i>	65
Issuer Participation in MARF.....	66
Appendix D: India IVR Transactions (SecureTelephone).....	67
Overview.....	68
Data Extensions to the Existing 3-D Secure Protocol.....	68
UCAF Transport in Mastercard Authorization Messages.....	68
Mastercard <i>SecureCode</i> —Security Level Indicator (DE 48, subelement 42).....	69
Universal Cardholder Authentication Field (DE 48, subelement 43).....	70
What is an AAV?.....	70
Sample IVR Transaction Flow.....	71
Mastercard <i>SecureCode</i> Compliance and Functional Testing.....	71
Appendix E: Mastercard Extensions for the Brazil Market.....	72
Brazil Market Extensions.....	73
Notices.....	7 5

Chapter 1 SecureCode Acquirer Implementation Overview

This section provides a general overview of the Mastercard® SecureCode™ Electronic Commerce program.

Mastercard and E-commerce.....	7
Maestro and E-commerce.....	7
Mastercard Debit.....	8
Mastercard SecureCode Program.....	8
Typical Payment Transaction Participants.....	10
Mastercard SecureCode Program Platform.....	11
What is UCAF and Its Structure?.....	11
UCAF Transport in Mastercard Authorization Messages.....	13
Sample DE 48 Contents.....	14
What is an AAV?.....	15
What is a Merchant Plug-In.....	16
Rules to Support UCAF Liability Shift.....	16
Liability Shift Rule Summary.....	17
UCAF Compliance Requirements.....	17
Authorization Requirements.....	17
Debit/Single Message System Requirements.....	20
Clearing Requirements.....	20
Enrollment Requirements.....	20
Maintenance Requirements.....	21
Participation Responsibilities for all Customers.....	21
Acquirer Responsibilities.....	22
Mastercard SecureCode Compliance and Functional Testing.....	23
Mastercard SecureCode Program Enrollment Procedure.....	23
Mastercard Support.....	24
Additional References.....	24

Mastercard and E-commerce

E-commerce transactions account for a significant and increasing share of Mastercard® gross dollar volume.

The number of remote transactions is increasing at a rate of more than 40 percent per year and growing. For this reason, it is important to position e-commerce and mobile commerce channels—web access from PCs, PDAs, mobile phones, and other wireless-enabled devices—to increase gross dollar volume profitability by using security and authentication solutions that authenticate cardholders. This reduces chargebacks and expenses that are associated with disputed transactions.

From a risk perspective, the current Mastercard electronic and mobile transaction environment closely resembles traditional mail order/telephone order (MO/TO) transactions. The remote nature of these transactions increases risk, resulting in more cardholder disputes, and associated chargebacks.

These factors increase costs to all parties for managing disputes and chargebacks. According to Mastercard data, more than 70 percent of all chargebacks for e-commerce transactions are associated with reason code 4837 (No Cardholder Authorization) or reason code 4863 (Cardholder Not Recognized), and are currently estimated at a cost of USD 34.00 per chargeback to the industry. These reason codes are used where the consumer denies responsibility for the transaction and the acquirer lacks evidence of the cardholder's authentication, or the consumer does not recognize the transaction.

Proving that the cardholder conducted and authorized the transaction in a virtual, non-face-to-face environment of electronic and mobile commerce has been extremely difficult. The Mastercard® *SecureCode*™ program is designed to provide the infrastructure for an issuer security solution that reduces problems associated with disputed charges, offering the opportunity to authenticate the cardholder at the time of purchase. Disputed charges affect all parties in a transaction—issuer, acquirer, cardholder, and merchant.

Maestro and E-commerce

Low credit card penetration in many countries has led to the use of inefficient payment forms like cash on delivery, check, and domestic transfer/automated clearing house (ACH).

Mastercard® *SecureCode*™ will allow Maestro® cards to be used for Internet purchases in a safe and secure environment. Mastercard *SecureCode* allows Maestro to be the first fully-authenticated global debit brand accepted on the Internet.

Unless otherwise stated by domestic country rules, all Maestro Internet transactions provide protection against chargebacks and fraud if the transaction was authorized by the issuer.

Unless otherwise stated by domestic country rules, all Maestro Internet transactions are guaranteed. Please note that a merchant can not accept Maestro transactions unless they support Mastercard *SecureCode*.

Mastercard rules require any merchant that wants to accept Maestro to undertake Mastercard *SecureCode* on every Maestro transaction unless they have applied and been accepted to the Maestro Advance Registration Program (MARF).

NOTE: MARF is closed to new business and will be decommissioned 1 June 2015. For more details, refer to *Europe Region Operations Bulletin No. 2*, 3 February 2014.

Mastercard recommends that issuers deploy an authentication solution for cardholders to have appropriate evidence of cardholder participation in the case of a disputed transaction.

Mastercard Debit

Issuers of Debit® Mastercard (either single or dual message) should note that the Single Message System is capable of supporting transport of the Universal Cardholder Authentication Field™ (UCAF) value.

Issuers must ensure that authorization, and clearing for dual message are compliant with the latest debit standards available on the Publications product on Mastercard Connect™. The Mastercard® *SecureCode*™ authentication program occurs before an authorization takes place, therefore, it can be used to authenticate any Mastercard card. Issuers should ensure that their systems can support the transport of the Account Authentication Value (AAV) before proceeding with a Mastercard *SecureCode* program.

Mastercard *SecureCode* Program

Mastercard® *SecureCode*™ offers flexible, robust, and easy-to-implement solutions for cardholder authentication.

Because requirements vary from issuer to issuer, Mastercard places a premium on flexibility, enabling issuers (with their ACS service provider) to choose from a broad array of security solutions for authenticating their cardholders. As the need for strong authentication and better consumer experience grow, new methods of authentication have evolved. The migration to dynamic authentication with risk-based solutions is becoming more common. Dynamic authentication can be one-time passwords (OTPs) via SMS, mobile device, or EMV chip with or without display cards.

From a program perspective there are a number of decisions that an issuer has to make prior to implementing Mastercard *SecureCode*.

- What is the issuer's attitude to risk?
- When should you authenticate a transaction? Do you want to authenticate every transaction or just those that fall into a certain risk profile? The *Mastercard Risk Based Authentication* white paper may be relevant if a percentage is preferred.
- How should you authenticate? See Issuer Authentication Options for a selection of possible Authentication options.
- How does this authentication strategy impact your current authorization strategy?

The decision on authentication strategy will influence the issuer's Mastercard SecureCode deployment and is, perhaps, the most important decision. This decision will balance the following considerations:

- The cardholder experience that an issuer wants to support for their cardholders. Just as card portfolios are segmented, authentication strategies can also be developed offering multiple forms of authentication (for example, SMS for youth, and display card for High Value).
- Cost of authentication
- Cost of fraud being experienced and projected
- Requirements from domestic regulators

When determining cost of deployment it is important to factor in the cost of both "Know Your Customer" and "Identification and Verification" that affects any opt in service deployed compared to the costs of a dynamic approach that does not require an opt-in by the cardholder.

From a program perspective there are several decisions that an acquirer has to make prior to implementing Mastercard SecureCode.

- Does the acquirer have concerns surrounding disintermediation or is it acceptable to allow third-party vendors and service providers to supply merchants solutions?
- Does the acquirer want to directly support merchants? If so, then this will require a minimum of a server based Merchant Plug-In solution available from any certified MPI vendor. A list of the certified MPI vendors is available at www.mastercard.us/merchants/securecode-vendors.html. Some merchants may require more than access to an MPI, including for example, fraud management, or web design, that are commonly provided by service providers such as DataCash. It is the acquirers responsibility to research what their customer's need and decide what propositions they want to offer.
- If reliant on third-party service providers who do you partner with? Are they certified for submission of authorization on and clearing to the acquirers host systems? Are they PCI-DSS compliant?

Although not within the acquirers area of responsibility, acquirers may be interested in the following authentication information.

The most common solutions for Mastercard and Maestro® issuers is the use of static or dynamic passwords. Dynamic password usage is typically based on the Chip Authentication Protocol (CAP) that provides for the creation of a one-time use cardholder authentication password. This scenario is similar to what the cardholder experiences in the face-to-face environment using EMV chip card and personal identification number (PIN). CAP is now available in a number of form factors ensuring that a single CAP Issuer solution can support SMS, Mobile Applications, Display Card and the traditional CAP models using sleeve devices. See Issuer Authentication Options for more information on authentication options. This program provides a seamless integration of both EMV and 3-D Secure technologies that result in stronger authentication than traditional static password solutions.

Mastercard SecureCode is the consumer-and-merchant-facing name for all existing and new Mastercard cardholder authentication solutions. While these solutions may each appear quite

different on the surface, these various approaches converge around the Universal Cardholder Authentication Field™ (UCAF) mechanism and share a number of common features.

In every case, for example, two things occur:

1. Mastercard or Maestro cardholders are authenticated using a secure, private code that is unique to them. Authentication can occur by weak implementation with static passwords; or a stronger authentication approach can be implemented where the issuer pre-enrolls cardholders with a static value or implements a one-time password (OTP) service provided through text message, chip card reader, or on an EMV display card. Some local regulators have more stringent rules based on region or country.
2. The authentication data is transported from party-to-party via the Mastercard UCAF mechanism.

For more information about the Mastercard *SecureCode* platform components, see the Mastercard *SecureCode* Program Platform section in this chapter.

Typical Payment Transaction Participants

In a face-to-face retail transaction or MO/TO transaction, electronic processing begins with the merchant or the acquirer. In a Mastercard® *SecureCode*™ transaction, the electronic processing begins with the cardholder.

The following paragraphs describe the bankcard payment participants that are involved in a Mastercard *SecureCode* transaction.

Cardholder

A cardholder is an authorized user of a card issued by a licensed financial institution. In the e-commerce environment, consumers and corporate purchasers interact with merchants remotely, typically through personal computers. However, these points of interaction are expanding to include new devices such as mobile phones. A cardholder uses a payment card issued by a licensed Mastercard financial institution. Mastercard *SecureCode* will ensure that a cardholder's identity is authenticated prior to the completion of a purchase from a participating merchant.

Cardholders participating in Mastercard *SecureCode* are expected to feel more confident when shopping online and being authenticated by their issuer.

Issuer

An issuer is a Mastercard financial institution customer that issues either only Mastercard cards or Mastercard and Maestro® cards. The issuer guarantees payment for an authorized transaction using the payment card in accordance with payment card brand regulations and local legislation. Specifically for Mastercard *SecureCode*, the issuer is responsible for providing authentication services for eligible online purchases.

Mastercard and Maestro issuers can benefit from participating in Mastercard *SecureCode* by increased transaction volume and value as consumer confidence increases from cardholder

authentication programs. Mastercard issuers can benefit from lower dispute management and processing costs.

Merchant

A merchant is a retailer, or any other person, firm, or corporation that, pursuant to a merchant agreement, agrees to accept Mastercard and/or Maestro cards when properly presented. With Mastercard *SecureCode*, the merchant can offer a cardholder an authenticated electronic interaction over the Internet. A merchant that accepts payment cards must have a relationship with an acquirer. Merchants can benefit from participating in Mastercard *SecureCode* in several ways including reduced fraud and dispute costs, increased transaction volume and value, protection from unauthorized card use, and access to the Maestro card base.

Acquirer

The Mastercard *SecureCode* program platform is comprised of a number of layered components. As described in the following pages, each of the components provides for specific authorization and authentication functionality during the processing of a Mastercard *SecureCode* transaction.

Mastercard *SecureCode* Program Platform

The Mastercard® *SecureCode*™ program platform is comprised of a number of layered components.

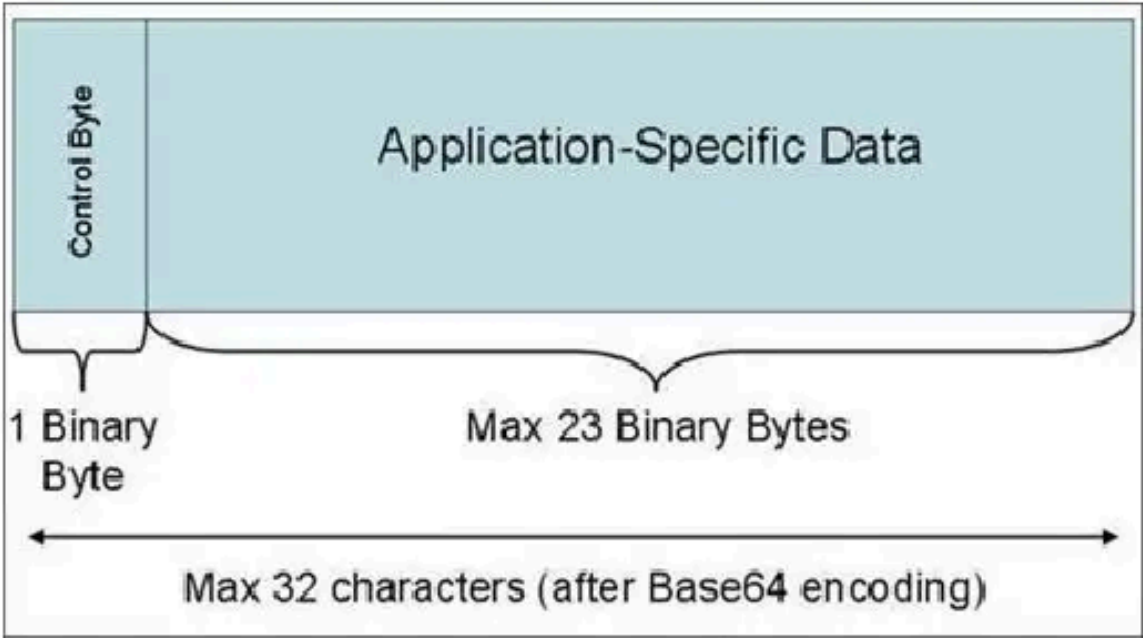
As described in the following sections, each of the components provides for specific authorization and authentication functionality during the processing of a Mastercard *SecureCode* transaction. When combined, the platform provides a mechanism for online merchants to receive a similar global payment guarantee to one that brick-and-mortar retailers enjoy with POS transactions.

What is UCAF and Its Structure?

Universal Cardholder Authentication Field™ (UCAF) is a standard, globally interoperable method of collecting cardholder authentication data at the point of interaction across all channels, including the Internet and mobile devices. This is also known as the Accountholder Authentication Value (AAV).

Within the Mastercard authorization networks (that is the Single Message and Dual Message System, and RSC) UCAF is a universal, multi-purpose data transport infrastructure that is used to communicate authentication information among cardholder, issuer, merchant, and acquirer communities. It is a variable length, 32-position field with a flexible data structure that can be tailored to support the needs of a variety of issuer security and authentication approaches.

The generic structure of UCAF is illustrated as follows:



The control byte contains a value that is specific to each security application. Mastercard is responsible for assigning and managing UCAF control byte values and the structure of UCAF application-specific data. Other solutions that use UCAF for authentication collection and transport will be assigned their own control byte value and the structure of the application-specific data will be tailored to support the specifics of the security protocol.

In most UCAF implementations, the application-specific data is defined as binary data with a maximum length of 24 binary bytes including the control byte. However, there are some restrictions in the various Mastercard authorization networks regarding the passing of binary data in the authorization messages. As a result, all UCAF data generated by Secure Payment Application™ (SPA) algorithm-based Mastercard® SecureCode™ implementations must be Base64 encoded at some point prior to being included in the authorization message. The purpose of this encoding is to produce a character representation that is approximately 33 percent larger than the binary equivalent. For this reason, the UCAF field is defined with a maximum length of 32 positions. For more information about Base64 coding, refer to the [#unique_12](#) appendix.

The current Mastercard SecureCode control byte definitions include the following.

Usage	Base64 Encoded Value	Hexadecimal Value
3-D Secure SPA Accountholder Authentication Value (AAV) for first and subsequent transactions	j	x'8C'
3-D Secure SPA AAV for attempts	h	x'86'

UCAF Transport in Mastercard Authorization Messages

Mastercard has designated specific subelements within data element (DE) 48 (Additional Data—Private Use) for the transport of Universal Cardholder Authentication Field™ (UCAF)-related data and associated indicators in authorization messages.

NOTE: UCAF data or an AAV may not be reused in authorization. It should only be sent with the original purchase authorization request. It should not be included in the Account status check inquiry or within subsequent recurring transactions.

The following sections describe these subelements.

DE 48, Subelement 42 (Electronic Commerce Security Level Indicator)

The Electronic Commerce Security Level Indicator contains the fields representing the security protocol and cardholder authentication type associated with the transaction. Mastercard requires that subelement 42 be included, and properly populated for all e-commerce transaction authorizations.

The preferred data values for Mastercard® SecureCode™ are:

Position	Value	Description
1—Security Protocol	2	Channel Encryption (for example, SSL)
2—Cardholder Authentication	1	Cardholder Certificate Not Used
3—UCAF Collection Indicator	0	UCAF data collection is not supported by the merchant or a Mastercard SecureCode merchant has chosen not to undertake Mastercard SecureCode on this transaction.
	1	UCAF data collection is supported by the merchant, and UCAF data is populated (DE 48, SE 43 is present) and contains an “attempt” AAV.)
	2	UCAF data collection is supported by the merchant, cardholder authentication was successful, and UCAF data is present and contains a fully authenticated AAV. (DE 48, SE 43)
	3	Merchant participates in MARP, Data collection is supported and UCAF data (the MARP static AAV) is present (DE 48, SE 43)

DE 48, subelement 43 (Universal Cardholder Authentication Field)

The UCAF contains the encoded result of the authentication. This data is collected by the merchant and may contain fully authenticated or attempts values generated by the issuing Access Control Server (ACS) or the Mastercard Attempts Processing Service.

NOTE: MARP is closed to any new business and was decommissioned 1 June 2015. For more details, refer to the *Europe Region Operations Bulletin No. 2*, 3 February 2014.

NOTE: The Universal Cardholder Authentication Field™ (UCAF) length is variable up to a maximum of 32 positions. All acquirers and issuers must ensure that they can send and/or receive contents in this field up to the maximum length of 32. Applications using this field, such as Mastercard SecureCode, may provide contents that are less than the maximum length. For Mastercard SecureCode specifically, this field will be 28 positions.

For additional information about the contents of this field for a Mastercard SecureCode authorization request message, see What is AAV? in this chapter.

Sample DE 48 Contents

Following are samples of different DE 48 contents.

Properly Coded DE 48 UCAF Fields

The following are examples of properly coded Universal Cardholder Authentication Fields™ (UCAFs) within a Mastercard® SecureCode™ authorization request message. The content of the DE 48, subelement 42 is underlined.

Scenario	Sample DE 48
Mastercard Advance Registration Program (MARP)	5MARPPROGRAM9999999999999999
Fully authenticated authorization	T42070103 <u>21</u> 24328jHyn+7YFi1EUAREAAAANUe6Hv8=820252 NOTE: The UCAF field (DE 48, subelement 43) is a variable length field up to a maximum of 32 positions. The Mastercard SecureCode Accountholder Authentication Value (AAV), in this case, is 28 characters in length. There must be no padding or trailing spaces in the UCAF field.
Unauthenticated authorization from UCAF enabled merchant containing an Attempt AAV	T42070103 <u>21</u> 14328hHyn+7YFi1EUAREAAAANUe6Hv8=820252
Unauthenticated authorization from non-UCAF enabled merchant	T42070103 <u>21</u> 0820252

Improperly Coded DE 48 UCAF Fields

The following are examples of improperly coded UCAF fields within a Mastercard SecureCode authorization request message. The content of the DE 48, subelement 42 is underlined.

Scenario	Sample DE 48
UCAF data field is improperly padded	T420701032124332jJLtQa+lws8AREAEbjsA1MAAAA= 820252. NOTE: The UCAF field (SE 43) is a variable length field up to a maximum of 32 positions. The Mastercard SecureCode AAV is 28 characters in length. There must be no padding or trailing spaces in the UCAF field.
Unknown Control Byte and invalid length indicator	T4207010321243328CC744DB69D5739CFE0100008EA89433820252 NOTE: The contents of the UCAF data field is not properly Base64. Instead the field contains a character representation of the hex Mastercard SecureCode AAV. The first character must be one of the documented control bytes.
UCAF data field contains binary AAV data instead of Base64 encoded data	T43328CC744DB69D5739CFE0100008EA89433820252 NOTE: The Security Level Indicator (DE 48, SE 42) is required when UCAF data (DE 48, SE 43) is present in the message. This results in a Banknet format error. Also, the UCAF data content is hex-encoded and not Base64-encoded.

What is an AAV?

The Accountholder Authentication Value (AAV) is a Mastercard® SecureCode™ specific token that uses the Universal Cardholder Authentication Field™ (UCAF) field for transport within Mastercard authorization messages.

It is generated by the issuer and presented to the merchant for placement in the authorization request. This AAV can be proof of a fully authenticated or an attempted authentication transaction.

The AAV is used to identify the processing parameters associated with the transaction. Among other things, the field values will identify the:

- Issuer ACS that created the AAV. (This could be the Issuer ACS or, in the case of an attempt, the Mastercard Attempt processing server.)
- Sequence number that can positively identify the transaction for that location
- Secret key used to create the Message Authentication Code (MAC), which is a cryptographic method that ensures AAV data integrity, and binds the entire AAV structure to a specific PAN.

UCAF is the mechanism that is used to transmit the AAV from the merchant to issuer for authentication purposes during the authorization process.

MARP Exception

For merchants that have qualified to participate in the Mastercard Advance Registration Program (MARP), Mastercard assigns a static AAV that conforms to the same format as an issuer-generated AAV.

NOTE: MARP is closed to new business and will be decommissioned 1 June 2015. For more details, refer to *Europe Region Operations Bulletin No. 2*, 3 February 2014.

What is a Merchant Plug-In

A merchant plug-in is a software application that is developed and tested to be compliant with the 3-D Secure protocol and interoperable with the Mastercard® SecureCode™ infrastructure.

The plug-in application is typically provided by a technology vendor and integrated with the merchant's commerce server. It serves as the controlling application for the processing of 3-D Secure messages.

As part of the Mastercard SecureCode infrastructure requirements, all merchant endpoints must implement application software capable of processing 3-D Secure messages. An endpoint is described as any merchant or merchant processor platform, which directly connects to the Mastercard SecureCode infrastructure.

NOTE: If a retailer has qualified and accepted a merchant for the Mastercard Advance Registration Program (MARP), then Mastercard will assign a static Accountholder Authentication Value (AAV) for use when the transaction is undertaken as MARP instead of standard Mastercard SecureCode. This value is passed in plain text in the Universal Cardholder Authentication Field™ (UCAF) field. For additional information, refer to the [Mastercard Advance Registration Program Requirements](#) appendix.

Rules to Support UCAF Liability Shift

In support of the Mastercard® SecureCode™ program, Mastercard has implemented liability shift programs governing specific fraud-related chargeback reason codes.

This section provides an overview of the Universal Cardholder Authentication Field™ (UCAF) liability shift programs in effect as of the date of publication. For more information, see the *Chargeback Guide*.

E-commerce Rules are provided in the *Transaction Processing Rules* available on Mastercard Connect®. Issuers should reference this publication for full details of the responsibilities and requirements when undertaking e-commerce transactions and the for the benefits of Mastercard SecureCode with this product.

DEFINITION: An intraregional transaction is one in which the acquiring institution and issuing institution are in the same Mastercard region, for example, the U.S., Canada, Europe, Asia Pacific, Middle East/Africa, and the Latin America/Caribbean regions.

An interregional transaction is one in which the acquiring institution and the issuing institution are in different Mastercard regions.

Liability Shift Rule Summary

Issuers should always check the *Chargeback Guide* for the current liability shift rules.

The following table summarizes the applicability of the liability shift rules for qualifying transactions across the Mastercard regions.

Liability Shift—Consumer Cards and Commercial Cards

* = Merchant-Only and Fully Authenticated Liability Shift

		Issuing Regions					
		U.S.	CAN	EUR	LA/C	AP	MEA
Acquiring Regions	U.S.	*	*	*	*	*	*
	CAN	*	*	*	*	*	*
	EUR	*	*	*	*	*	*
	LA/C	*	*	*	*	*	*
	AP	*	*	*	*	*	*
	MEA	*	*	*	*	*	*

UCAF Compliance Requirements

Support for processing of Universal Cardholder Authentication Field™ (UCAF) transaction data is a co-requisite for all Mastercard® SecureCode™ acquirer implementations.

Authorization Requirements

Mastercard issuers and acquirers must modify their authorization systems to support the required Universal Cardholder Authentication Field™ (UCAF) transaction data elements.

Further, it is a requirement that all qualifying transactions must be properly coded e-commerce transactions.

The following table highlights the field requirements that are required for coding of Mastercard® SecureCode™ e-commerce authorizations. For more information about the

coding of Mastercard SecureCode UCAF data elements for non-e-commerce transactions, refer to the appropriate authorization message specification document. When implementing Mastercard SecureCode with Masterpass™ wallet, refer to the *Masterpass Merchant Implementation Guide* for authorization processing rules. When implementing SecureCode with Masterpass wallet, please refer to the Masterpass Merchant implementation guide for authorization processing rules.

Data Element		Comments
22	Point of Service Entry Mode	
	Subfield 01—POS Terminal Entry Mode	81—PAN entry via electronic commerce, including chip
48	Additional Data—Private Use	
	TCC—Transaction Category Code	Contains one of the following values: • P = Payment Transaction • T = All other non-face-to-face transactions • U = Unique (Refer to the <i>GCMS Reference Manual</i> for additional requirements regarding the use of this value. • X = Airline and Other Transportation Services (regardless of whether or not the transaction originates at a face-to-face terminal)
Subfields		
	42—Electronic Commerce Security Level Indicator (Mandatory)	Contains the security level in positions 1 and 2 and the UCAF collection indicator in position 3 • Position 1 = Security Protocol (for Mastercard SecureCode, the value is 2—Channel) • Position 2 = Cardholder Authentication (for Mastercard SecureCode, the value is 1—Cardholder Certificate Not Used)

Data Element	Comments
42—Electronic Commerce Security Level Indicator (Mandatory)	Position 3—UCAF Collection Indicator. Valid values are: 0 = UCAF data collection is not supported by the merchant for this transaction or a Mastercard <i>SecureCode</i> merchant has chosen not to undertake Mastercard <i>SecureCode</i> on this transaction. 1 = UCAF data collection is supported by the merchant, and UCAF data may be available (DE 48, subelement 43 should contain an “attempt” AAV for Mastercard <i>SecureCode</i>) 2 = UCAF data collection is supported by the merchant and UCAF data must be populated and contains a fully authenticated AAV. (DE 48, subelement 43) 3 = Merchant participates in MARP and UCAF must be populated with the Mastercard assigned static AAV (DE 48, sub element 43)
43—Universal Cardholder Authentication Field (Conditional)	Contains the encoded Mastercard <i>SecureCode</i> -generated authentication data.
61 Point-of-Service Data	Contains the following values to uniquely identify the ecommerce transaction.
Subfields	
3—POS Terminal Location	2 = Off premises of card acceptor facility (cardholder terminal including home PC, mobile phone, PDA)
4—POS Cardholder Presence	5 = Electronic order (home PC, Internet, mobile phone, PDA)
10—Cardholder Activated Terminal Level	6 = Authorized Level 6 CAT: Electronic commerce

Debit/Single Message System Requirements

Issuers and acquirers of Debit Mastercard and Maestro® cards must modify their authorization systems to support Universal Cardholder Authentication Field™ (UCAF) transaction data.

For more information about requirements to support UCAF, refer to the *Single Message System Specifications* manual.

Clearing Requirements

In support of the Mastercard® SecureCode™ program, Mastercard requires that acquiring members must support changes to the Global Clearing Management System (GCMS) clearing records in support of associated interchange rate programs.

Mastercard requires that all clearing and chargeback records associated with an e-commerce authorization message contain the DE 48, private data subelement 0052 (Electronic Commerce Security Level Indicator). Failure to include this field will result in the message being rejected by GCMS with an edit error. The contents of this field must be consistent with the value of the DE 48, subelement 42, position 3 in the corresponding authorization message. In addition, Universal Cardholder Authentication Field™ (UCAF) interchange rate qualifications for all transactions acquired in the U.S. region are subject to interchange compliance edits.

NOTE: Issuers should refer to the latest *Interchange Manual* for their region to determine what interchange is applied for Full UCAF and Merchant UCAF transactions.

Enrollment Requirements

Following are acquirer requirements to participate in the Mastercard® SecureCode™ Electronic Commerce Program:

- All participating customers must complete both a general enrollment form and the acquirer program enrollment form. Refer to the [Mastercard SecureCode Program Enrollment Procedure](#) section for additional information.
- Select a vendor or service provider to provide the required infrastructure (for example, merchant plug-ins, access control servers, and more) if the acquirer has decided to directly support an MPI.
- Select a vendor or service provider to provide the required infrastructure (for example, merchant plug-ins, access control servers, and more) if the acquirer has decided to directly support an MPI.
- Assign a program manager to manage, test, and integrate the current system with Mastercard SecureCode, or to manage the recruitment of third parties or service providers to provide Mastercard SecureCode to their e-commerce merchants.
- Ensure all authorization and clearing systems can support Universal Cardholder Authentication Field™ (UCAF) data. For additional information, refer to the [UCAF Compliance Requirements](#) section.

The Mastercard E-commerce Program Manager will review and approve customer enrollment materials.

Maintenance Requirements

Issuers and acquirers should regularly review the content of their existing registration document, especially when key individuals noted in the registration move to other roles or leave the organization.

If changes are required, update the form and send it for amendment of the current registration to securecode_customer_support@mastercard.com. Issuers should regularly maintain their Mastercard® SecureCode™ contact in the Member Information-Mastercard by accessing the Applications page from Mastercard Connect™.

Participation Responsibilities for all Customers

The following sections explain customer roles and responsibilities for participation in the Mastercard® SecureCode™ program. These roles and responsibilities are applicable for both Mastercard and Maestro® issuers.

Customer Agreement to Abide by Mastercard Rules and Policies

Mastercard agrees to permit a member to participate in transactions involving the Mastercard SecureCode program in return for which member agrees to be bound by all of the rules, policies, and procedures relating to the Mastercard SecureCode program in effect at the time the transactions are conducted whether formally enacted by Mastercard or communicated through brochures, statements, guidelines or other member communications. Without limiting the foregoing, member specifically acknowledges and agrees to be bound by the Mastercard implementation of Mastercard SecureCode. Member understands and acknowledges that all rules and policies relating to the Mastercard SecureCode program and electronic commerce may be amended from time to time and are subject to change without notice by Mastercard.

No Representations or Warranties

Mastercard makes no representations or warranties whatsoever with respect to the Mastercard SecureCode program, including without limitation any software, hardware or other technology built to Mastercard standards, any services provided by it, or any other matter relating to or arising out of any implementation of the Mastercard SecureCode specifications. Without limiting the foregoing, Mastercard specifically disclaims all representations and warranties as to its operation of the Mastercard Certificate Authority. Mastercard specifically disclaims all implied warranties of merchantability, fitness for a particular purpose and non-infringement of third party intellectual property rights with respect to any and all Mastercard SecureCode implementations whether or not Mastercard has been advised, has reason to know, or is otherwise in fact aware of any information regarding the foregoing.

Indemnity and Limitation of Liability

Member agrees to indemnify and hold harmless Mastercard and its directors, officers, employees, agents and affiliates against any and all claims arising out of member's participation in the Mastercard *SecureCode* program. Member agrees that it shall not assert any claim or action against Mastercard arising out or otherwise connected in any way with the Mastercard *SecureCode* program and hereby waives all such claims against Mastercard and releases Mastercard from and against any and all liability for such claims. Without limiting the foregoing, member specifically waives and releases Mastercard from any and all claims arising out of its operation of the Mastercard *SecureCode* certificate authority. The foregoing limitations of liability shall apply to any claim or cause of action under law or equity whatsoever, including contract, warranty, strict liability, or negligence, even if Mastercard has been notified of the possibility of such claims. The entire liability of Mastercard to member for any and all acts, errors, and omissions relating to or arising out of or in connection with the Mastercard *SecureCode* program shall be limited to USD 1,000.

License Agreement for Specifications

If member executes a license agreement for any Mastercard specifications related to the Mastercard *SecureCode* Program (whether by written, click-wrap, or shrink-wrap agreement), the terms of such license agreement shall have precedence and govern over the terms in the Participation Responsibilities section of this chapter.

Acceptance of Terms

The signing of enrollment forms indicates that the member understands and agrees to the terms and conditions set forth herein including but not limited to those above. The member acknowledges that its acceptance of these terms and conditions is relied upon by Mastercard in permitting member's participation in the Mastercard *SecureCode* program.

Acquirer Responsibilities

The following are responsibilities specific to acquirers participating in the Mastercard® *SecureCode*™ program.

- Soliciting merchants to participate in the Mastercard *SecureCode* program.
- Adhering to:
 - Mastercard requirements for information and data security, including but not limited to protection of Mastercard card numbers, Maestro account numbers, cardholder enrollment information and management and protection of secret cryptographic keys.
 - Applicable usage guidelines as published in the [Mastercard *SecureCode* Program Identifier Usage Guidelines](#) section.
- Ensuring that merchants comply with the requirements documented within the *Mastercard SecureCode—Merchant Implementation Guide*.
- Reporting to Mastercard on experiences and progress with merchants use of Mastercard *SecureCode* products, shopping experiences, and related market research information.
- Providing:

- Primary customer service for merchants.
- Testing resources to test the ability to process Authorization Request/0100 (for Mastercard and Maestro cards in the Europe region) and Authorization Request/0200 (for Maestro cards) messages and receive Authorization Request Response/0110 (for Mastercard cards and Maestro cards in the Europe region) and Authorization Request Response/0210 (for Maestro cards) messages containing data specific to e-commerce.

Mastercard SecureCode Compliance and Functional Testing

Mastercard has developed vendor compliance testing, as well as issuer and merchant functional testing, to ensure vendor products and member and merchant implementations are compliant and successfully interoperate with all Mastercard® SecureCode™ platforms and infrastructure.

Proof of Visa Certification of IVR-ACS and IVR-MPI development is required before Mastercard compliance testing can commence.

All vendors, merchant endpoints, and issuers are required to complete the designated testing process prior to launch. More information regarding testing is available from securecode_customer_support@mastercard.com

Mastercard SecureCode Program Enrollment Procedure

The following information outlines the Mastercard® SecureCode™ enrollment procedure.

1. Request enrollment forms by sending an email message to securecode_customer_support@mastercard.com
2. Complete and sign both the General and Acquirer enrollment forms for the Mastercard SecureCode Electronic Commerce Program.
3. Submit the enrollment forms by email message to securecode_customer_support@mastercard.com

After the authorized representative receives the completed and signed enrollment forms, the Mastercard E-commerce Program team will:

1. Review and approve the forms and forward them to the appropriate internal groups for review and approval. If necessary, site visits are scheduled.
2. Provide testing details and instructions to program contacts listed on the enrollment forms.

Mastercard Support

Mastercard will provide support to customers enrolled in the Mastercard® SecureCode™ program.

Project Management Support

Mastercard team members are available to provide guidance and advice about member implementation, subject to personnel availability. For specific contact information, refer to the [Mastercard SecureCode Contact Information](#) section.

Regional Mastercard Office Support

Regional Mastercard offices will provide direction and planning support subject to personnel availability.

All advice and information provided by Mastercard is subject to the disclaimers and limitations on liability.

Additional References

Unless otherwise noted, all publications are available from the Publications section of Mastercard Connect™.

Mastercard SecureCode Program Publications

Mastercard® SecureCode™ program publications include:

- Licensed publications available by email request to securecode_customer_support@mastercard.com
 - *SPA Algorithm for the Mastercard Implementation of 3-D Secure*
 - *SecureCode ACS Security Requirements*
 - *Mastercard SecureCode Production Procedures*
- The following, which are available by sending an email request to authentication@mastercard.com:
 - *Chip Authentication Program—Functional Architecture*
 - *Chip Authentication Program—Implementation Guide*
 - *CAP 2007 Specifications Updates*
 - *CAP 2007 Implementation Guide Update—CDOL*
- The Mastercard SecureCode Toolkit, which can be accessed via the following web address:
 - www.mastercard.com/us/merchant/security/what_can_do/SecureCode/toolkit/SecureCode.html

NOTE: The documentation regarding the industry standard for 3-D Secure is not available from Mastercard. These documents must be sourced under license from the standard owner.

Mastercard Authorization Specification Publications

Mastercard Authorization specification publications are as follows:

- *Customer Interface Specification*
- *Regional Service Center Message Specification Manuals*
- *Single Message Specifications*
- *Quick Reference Booklet* (contains state, country, and currency codes)
- Mastercard SecureCode web training and information presentations are available at <https://globalrisk.mastercard.com/securecode-360/>.
- White paper on Risk-based Authentication is available at <https://globalrisk.mastercard.com/online-resources/>.
- Issuer Best Practice Guide for SecureCode available at <https://globalrisk.mastercard.com/online-resources/>.

Additional Mastercard Publications

Additional Mastercard publications that may be helpful include:

- *Mastercard SecureCode—Issuer Implementation Guide*
- *Mastercard SecureCode—Merchant Implementation Guide*
- *GCMS Reference Manual*
- *Transaction Processing Rules*
- *Interchange Manual* (region-specific)

3-D Secure Protocol Specification Publications

3-D Secure Protocol Specification documents include:

- 3-D Secure Functional Requirements Merchant Server Plug-In v1.0.2 6 May 2006
- 3-D Secure Functional Requirements Access Control Server v1.0.2 6 May 2006

Chapter 2 3-D Secure Solution

This section provides an overview of the Mastercard implementation of 3-D Secure for Mastercard® cards and Maestro® cards, including cardholder enrollment and payer authentication.

3-D Secure Solution Overview.....	27
Components.....	27
Issuer Domain.....	27
Acquirer Domain.....	29
Interoperability Domain.....	29
3-D Secure Solution Message Types.....	31
Card Range Request/Response.....	31
Verification Request/Response.....	31
Payer Authentication Request/Response.....	32
Payer Authentication Transaction Request/Response.....	32
Cardholder Enrollment.....	32
Cardholder Enrollment Process.....	32
Cardholder Authentication.....	33
Sample Cardholder Authentication Process.....	33

3-D Secure Solution Overview

Cardholder authentication is the process of verifying cardholder account ownership during a purchase transaction in an online electronic commerce environment.

All Mastercard® *SecureCode*™ solutions define and provide a base level of security around performing this authentication process. For this solution specifically, Mastercard is deploying its own implementation of 3-D Secure under the Mastercard *SecureCode* program branding for Mastercard® and Maestro®. This implementation of 3-D Secure includes support for the Secure Payment Application™ (SPA) algorithm and Universal Cardholder Authentication Field™ (UCAF) without any changes to the 3-D Secure specification, messages, or protocol.

The components described herein are organized according to requirements that fall within the issuer, acquirer, and interoperability domains associated with the payment process.

- Issuer Domain—Systems and functions of the card issuing financial institutions and its customers.
 - Cardholder Browser
 - Related Cardholder Software
 - Enrollment Server
 - Access Control Server
 - Accountholder Authentication Value (AAV) Validation Server/Process
- Acquirer Domain—Systems and functions of the acquirer and its customers.
 - Merchant Plug-In
 - Signature Validation Server
- Interoperability Domain—Systems, functions, and messages that allow the Issuer Domain and Acquirer Domain to interoperate. These components will be globally operated and managed by Mastercard.
 - Directory Server
 - Certificate Authority
 - Mastercard Authentication History Server
 - Attempts Processing Server

Components

Following is information about components related to the Issuer Domain, Acquirer Domain, and Interoperability Domain.

Issuer Domain

The Issuer Domain is comprised of the following components.

- Cardholder browser and related software
- Enrollment server

- Access control server
- Accountholder Authentication Value (AAV) validation server/process

Cardholder Browser and Related Cardholder Software

The Cardholder browser acts as a conduit to transport messages between the Merchant Server Plug-In (in the Acquirer Domain) and the Access Control Server (in the Issuer Domain). Optional cardholder software to support implementations such as chip cards may also be included.

Both the browser and related software are considered to be off-the-shelf components that do not require any specific modification to support 3-D Secure.

Enrollment Server

The purpose of the enrollment server is to facilitate the process of cardholder enrollment for an issuer's implementation of 3-D Secure under the Mastercard® *SecureCode*™ program. The server will be used to perform initial cardholder authentication, as well as administrative activities such as *SecureCode* resets and viewing 3-D Secure payment history. In some cases, the enrollment server and the access control server may be packaged together.

Access Control Server

The access control server serves two basic, yet vital, functions during the course of a Mastercard *SecureCode* online purchase. First, it will verify whether a given account number is enrolled in the Mastercard *SecureCode* program. Secondly, it will facilitate the actual cardholder authentication process.

AAV Validation Server/Process

This server or process will be used to perform validation of the cardholder authentication data received by the issuer's authorization system in the authorization messages. Issuers may perform their own validation or sign up for the Mastercard-hosted on-behalf service. To register for the Mastercard-hosted on-behalf service, send an email to your regional Customer Implementation Service team.

Mastercard recommends that issuers validate the AAV contained in the authorization message prior to the authorization decision. This is considered a best practice, although it is not required. With the implementation of the Mastercard Attempts Processing Service (Stand-In Authentication), issuers that perform self-AAV validation will continue to be able to validate AAV values generated by their Access Control Server (ACS) service. However, an issuer will not have the ability to validate an attempt AAV generated by the Mastercard Attempts Processing Service. Issuers that participate in the AAV validation on-behalf service will have all AAV values validated.

The server of the Mastercard Attempts Processing Service will generate an attempt AAV when:

- The Issuer's account ranges do not participate in Mastercard *SecureCode*.
- The Issuer's ACS responds negative to the merchant enrollment verification message (VERes = N) for unenrolled cardholders.

- The Issuer's ACS times out or is unavailable.

Under these circumstances, an issuer will not be able to validate a Mastercard generated attempts AAV. For security reasons, the keys will not be shared.

For details about the AAV validation on-behalf service, refer to the Authorization Services Details chapter of the *Authorization Manual*.

Acquirer Domain

The Acquirer Domain is comprised of the following components.

- Merchant plug-in (MPI)
- Signature validation server

Merchant Plug-In

The merchant server plug-in creates and processes payer authentication messages and then returns control to the merchant software for further authorization processing. The plug-in is invoked after the cardholder finalizes the purchase request, which includes selecting the account number to be used, and submitting the order but prior to obtaining authorization for the purchase.

Signature Validation Server

The signature validation server is used to validate the digital signature on purchase requests that have been successfully authenticated by the issuer. This server may be integrated with the merchant plug-in or may be a separately installed component.

Interoperability Domain

The Interoperability Domain is comprised of the following components.

- Directory server
- Certificate authority
- Mastercard Authentication History Server
- Attempts server

Directory Server

The Mastercard[®] *SecureCode*[™] global directory server provides centralized decision-making capabilities to merchants enrolled in the Mastercard *SecureCode* program. Based on the account number contained in the merchant enrollment verification request message, the directory will first determine whether the account number is part of a participating Mastercard or Maestro[®] issuer's card range. It will then direct eligible requests to the appropriate access control server for further processing.

All implementations of this platform **must** use the Mastercard *SecureCode* global directory server for processing Mastercard and Maestro[®] transactions.

Certificate Authority

The Mastercard Certificate Authority is used to generate and distribute all private hierarchy end-entity and subordinate certificates, as required, to the various components across all three domains.

These certificates include:

- Mastercard Root certificate (used for both Mastercard and Maestro)
- SSL Server and Client certificates issued under the Mastercard hierarchy for communication to the Directory Server and Mastercard Authentication History Server
- Issuer Digital Signing certificates issued under the Mastercard hierarchy for communication to the Directory Server and History Server

In addition, SSL certificates based on a public root hierarchy are required. These certificates are not issued by the Mastercard Certificate Authority and must be obtained from another commercially available certificate-issuing provider.

For more information, refer to the *Mastercard SecureCode—Production Procedures* manual.

Mastercard Authentication History Server

The Mastercard Authentication History Server is a central repository of all authentication activity occurring out of band from Mastercard Authorization systems. All Access Control Server (ACS) service providers should implement communications with the History Server. For questions relating to the History Server, contact Mastercard *SecureCode* Customer Support.

The History Server is a central repository of all authentication activity occurring within the issuer ACS for all transactions that occurred, including the PAREq and PAREs details.

The Mastercard *SecureCode* infrastructure supports this component server. All Access Control Server (ACS) providers will need to implement communication with the History Server. Send an email to securecode_customer_support@mastercard.com with questions.

Attempts Server

The Mastercard *SecureCode* infrastructure supports this component server.

The Mastercard Attempt processing server provides the merchant with an Attempt Accountholder Authentication Value (AAV) when:

- The Issuer account range is not participating on the Mastercard *SecureCode* Directory Server.
- The Issuer sends a verify enrollment response with the value of an N (Cardholder not enrolled).
- The Issuer ACS is Unavailable or times out.

Issuers may choose to have their ACS provider to send a VRes=Y for non-enrolled cardholders, and then providing the PAREs=A to avoid invoking the Mastercard Attempts Processing Service.

The AAV generated by the attempts Server service will be generated with Mastercard keys which will not be shared. An issuer can identify an AAV generated by the new Mastercard Attempts Processing Service by the ACS identifier in the base64 decoded version of the AAV.

Attempts Server

With the implementation of the Mastercard Attempts Processing Service's "Stand-In" Authentication, Attempts AAVs will be generated in the following cases:

- ACS Services are not responding to Verify Enrollments.
- Issuer card ranges are not participating or enrolled in the Directory Server.
- A cardholder is not enrolled or participating in the issuer's service.

Issuers that perform self-validation on the AAV at the time of authorization are not be able to validate an Attempts AAV generated by the Mastercard Attempts Processing Service since the key which generates this value by Mastercard will not be provided or shared.

3-D Secure Solution Message Types

The card range request/response, the verification request/response, the payer authentication request/response, and the payer authentication transaction request/response are all message types associated with the 3-D Secure Solution process.

Card Range Request/Response

Card Range Request/Response, also known as card range caching, is no longer a viable implementation. The Mastercard Attempts Processing Service generates a Stand-In Authentication for cardholders not enrolled, card ranges not participating, and Access Control Server (ACS) services not responding. This Attempts Accountholder Authentication Value (AAV) is only provided upon Merchant Plug-In (MPI) communication to the Directory Server for each transaction. This ensures proper processing by all parties.

Verification Request/Response

Message Pair: VReq/VERes— The first step in the payer authentication process is to validate issuer and cardholder participation in the 3-D Secure. As designed by the protocol for the safety and security of the authentication, the Mastercard Directory Server validates and routes the VReq/VERes between Merchant Plug-In (MPI) and Access Control Server (ACS) providers.

Therefore, the verification request process sent by the merchant is a two-step method resulting in two verification requests (VReqs).

1. A VReq is sent from the MPI to the Directory Server to check card range enrollment and routing to the appropriate authenticating ACS.
2. A second VReq is sent from the Directory Server to the appropriate ACS for cardholder enrollment verification and response back through the Directory Server.

All merchants will need to call the directory server for every Mastercard *SecureCode* transaction to receive an Attempts Accountholder Authentication Value (AAV) or an AAV from

the issuer's ACS provider. Upon implementation of the Attempts Service, MPI services will only see VERes message values of Y. The VERes will route the MPI service via the cardholder's browser to either the cardholder's ACS Service or the Mastercard Attempts Processing Service for Stand-In authentication.

Payer Authentication Request/Response

Message Pair: PAREq/PARes—After determining that a cardholder is enrolled to participate in 3-D Secure, the actual process of payer authentication is performed for each online purchase.

The Payer Authentication Request/Response messages are sent from the Merchant Server Plug-In to the Access Control Server to initiate the actual authentication. At this point in the process, cardholders will be presented with an authentication window and asked to enter their *SecureCode* or one-time password (OTP).

The Access Control Server (ACS) will perform authentication and, if successful, generate an Accountholder Authentication Value (AAV). It is returned to the merchant within the PARes message. For successfully authenticated transactions and Attempts, this AAV must be sent by the merchant to the acquirer and forwarded to the issuer as part of the authorization request. ACS providers should provide AAV values for all attempts (PARes = A) when the cardholder is not enrolled or declines activation in addition to the fully authenticated (PARes = Y) transaction status.

Payer Authentication Transaction Request/Response

Message Pair: PATransReq/PATransRes—Following authentication, it may be desirable to centralize storage of authentication requests for later dispute processing.

The Payer Authentication Transaction Request/Response messages provide a record of this authentication activity sent from the Access Control Server (ACS) to the Mastercard Authentication History Server. All ACS service providers must support the PATransReq/PATransRes messages for the History Server.

The Mastercard® *SecureCode*™ global infrastructure supports the History Server. All ACS providers must support these messages.

Cardholder Enrollment

Enrolling cardholders in the Mastercard Location Alerts is the sole responsibility of the issuer. This section outlines the enrollment requirements and process.

Cardholder Enrollment Process

The following section outlines the cardholder enrollment process for Mastercard® *SecureCode*™ when an opt-in service such as static password is used.

This will be impacted by the decision on which authentication has been taken, and if risk-based authentication has been implemented. Cardholder enrollment is not required for some of these dynamic/risk-based methods of authentication.

There are two primary types of interactions between a cardholder and the issuer software:

- Enrollment—if static password
- Purchase Transactions—when risk-based authentication does not occur and the cardholder needs to be challenged

Supplemental functionality is available based on Access Control Server (ACS) service provider functionality:

- Account Management—to view historical activity or other account profile information
- Reregistration—if static password is used or mobile phone collection is necessary for SMS
- Frequently Asked Questions (FAQs)
- Contact Details

To shop with the assurance of 3-D Secure, a cardholder must enroll in the program through the issuer. The enrollment process verifies the cardholder's identity and allows the cardholder to establish a *SecureCode* for authentication while shopping online. Enrollment of cardholders can be a weak implementation with static passwords after answering basic enrollment questions or a stronger authentication approach where the issuer pre-enrolls cardholders with a static value or implements a one-time password (OTP) service. These options can be discussed and reviewed with ACS Service Providers.

Cardholder Authentication

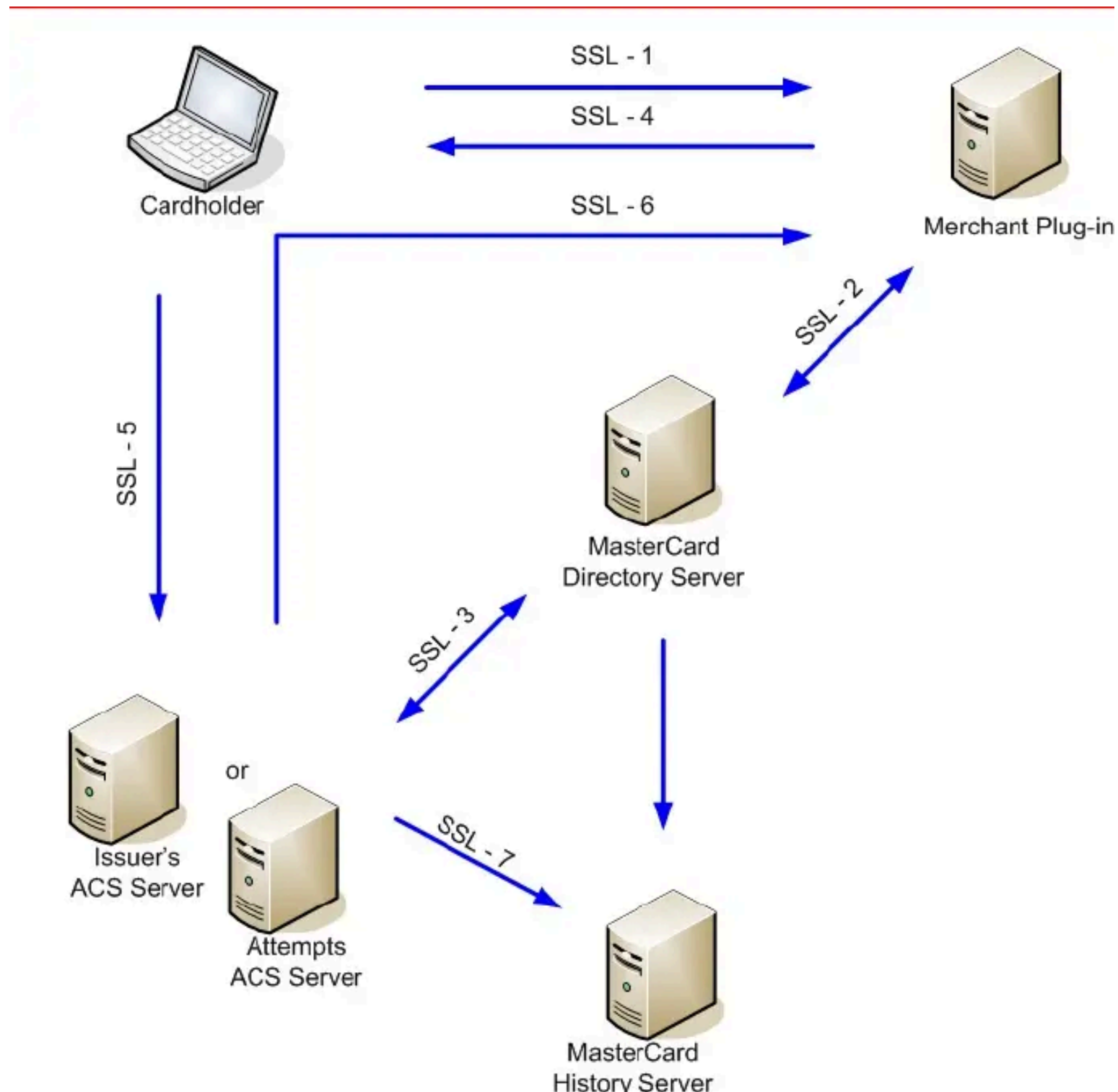
Following is information about the cardholder authentication process.

Sample Cardholder Authentication Process

The sample flow that follows assumes that the cardholder has already enrolled in the issuer's Mastercard® *SecureCode*™ program and obtained a *SecureCode* to use while shopping online at participating merchants.

The figure below also assumes that all communication channels between the various components are properly secured using the Secure Socket Layer (SSL) protocol.

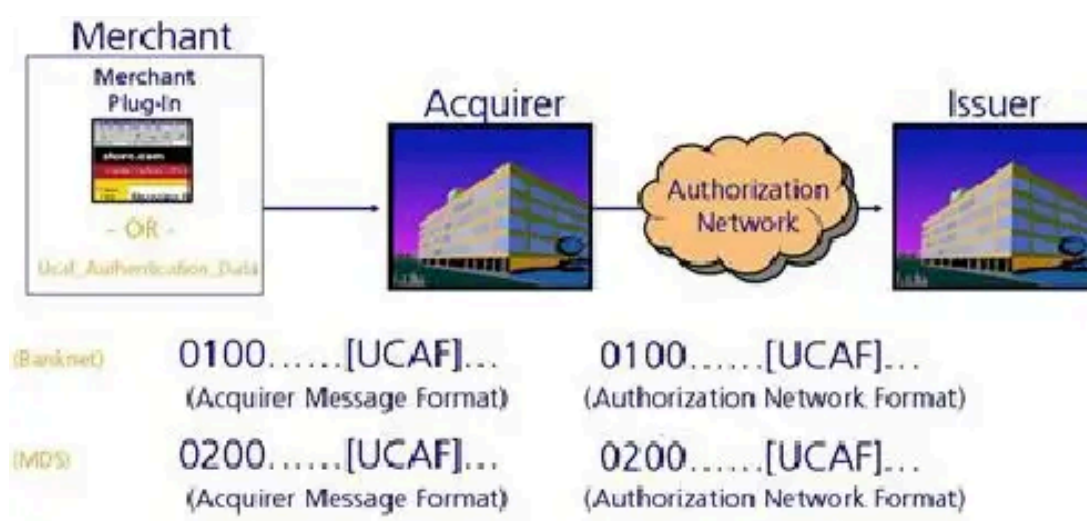
Refer to the [Components](#) section for additional information.



Link	Description
SSL-1	The cardholder shops at the merchant and, when ready to checkout, enters the appropriate payment information—including the account number.
SSL-2	The Merchant Plug-In queries the Directory to verify the enrollment status for a specific issuer using the verification request messages.

Link	Description
SSL-3	The Directory Server determines routing of the enrollment request to send a second VReq from the directory to the Access Control Server for enrollment verification and response to complete the cardholder authentication. The configuration information in the Directory will indicate exactly which Access Control Server will perform the check. The resulting response will flow back over the same links to the Merchant Plug-In.
SSL-4	The Merchant Plug-In creates the Payer Authentication Request message and sends it to the cardholder's browser.
SSL-5	The cardholder browser redirects the message to the appropriate Access Control Server to perform cardholder authentication. When the Access Control Server receives the Payer Authentication Request message, it causes the user authentication dialog to begin. This in turn causes a separate authentication window to appear to the cardholder that will facilitate the cardholder authentication process.
SSL-6	The Access Control Server authenticates the cardholder <i>SecureCode</i> , constructs the SPA AAV for Mastercard's implementation of 3-D Secure, and builds and digitally signs the Payer Authentication Response message. It is returned to the Merchant Plug-In, at which point the cardholder authentication window will disappear.
SSL-7	The Access Control Server sends PATransRec to the Mastercard Authentication History Server.

After cardholder authentication is complete, the merchant is required to pass the corresponding Secure Payment Application™ (SPA) Accountholder Authentication Value (AAV) to the acquirer via the Universal Cardholder Authentication Field™ (UCAF) within the authorization message. This value is then passed from the acquirer to the issuer as part of the authorization message.



When received by the issuer, the AAV can be validated as part of authorization request processing, as well as archived for use in potential cardholder disputes. Issuers will only be able to validate an Attempt or fully authenticated AAV generated by their own ACS. If the

Mastercard Attempts Processing Service generates an AAV, it cannot be issuer validated. The Mastercard Attempts Processing Service uses a unique key that cannot be shared. For issuers wanting to ensure that all AAVs are validated, Mastercard offers the Mastercard-hosted on-behalf service. AAVs will also be generated by the Mastercard Attempts Processing Service for Stand-In Authentication when cardholders are not enrolled, card ranges are not participating in Mastercard *SecureCode*, or Access Control Server (ACS) services are not responding. This Attempts AAV can only validate through the Mastercard AAV Validation Service rather than through self-validation.

Chapter 3 Mastercard SecureCode Cardholders

This section provides an overview of the various activities and requirements for building and maintaining the cardholder components that are required to support Mastercard® SecureCode™.

Cardholder Overview.....	38
Cardholder Infrastructure.....	38
Cardholder Customization.....	38
Cardholder Operational.....	38
Accountholder Authentication Value Processing.....	39
Cardholder Global Infrastructure Testing Requirements.....	39

Cardholder Overview

This section outlines the major activities and requirements for building and maintaining the cardholder components that are required to support Mastercard® *SecureCode*™.

The activities and requirements are separated into five primary categories:

- Infrastructure—Installation of new hardware and software components.
- Customization—Customizing or configuring vendor products.
- Operational—Operating the components in a production environment, including any process oriented changes that may be required.
- Accountholder Authentication Value (AAV) Processing—Handling and processing of the AAV.
- Global Infrastructure Testing Requirements—Testing of Mastercard *SecureCode* platform components with the Mastercard *SecureCode* global infrastructure.

Cardholder Infrastructure

Following is the cardholder infrastructure requirement for the installation of new hardware and software components that support Mastercard® *SecureCode*™.

Availability of Secure Browser

Mastercard recommends that cardholders have a web browser that is capable of supporting 128-bit Secure Sockets Layer (SSL) encryption.

Cardholder Customization

There are no customization issues related to cardholder participation in the Mastercard® *SecureCode*™ program.

Cardholder Operational

Following are the cardholder requirements for operating the components in a production environment, including any process-oriented changes that may be required.

Cardholder Registration

Depending on the exact nature of a card issuer's program, cardholders may be asked to register with their issuer to participate in its Mastercard® *SecureCode*™ program. Issuers have the option of allowing cardholders to enroll in the program while shopping at a participating merchant or outside of this process all together. The exact method of registration will be both issuer and issuer platform specific.

For accounts with more than one authorized user, both users should be able to register and create their own unique *SecureCode*. After one of the authorized card users has registered, all online payments using that card—at participating merchants—will cause the cardholder authentication process to initiate. This process will initiate even if the buyer is not the cardholder that explicitly enrolled in the program.

Use of Pop-Up Blocking Software

Early merchant implementations of Mastercard *SecureCode* used pop-up authentication windows. Pop-up blocking software used by cardholders inhibited these types of authentication windows from working properly. Mastercard requires that all merchant implementations use inline authentication windows because they are not impacted by the use of pop-up blocking software.

NOTE: Pop-up blocking software prevents most types of advertisement windows from being created or “popping up” while a user browses websites.

Accountholder Authentication Value Processing

There are no issues related to processing of the Secure Payment Application™ (SPA) Accountholder Authentication Value (AAV) for cardholders.

Cardholder Global Infrastructure Testing Requirements

There are no testing requirements for cardholders.

Chapter 4 Customer Interface Requirements

This section helps customer personnel understand the Mastercard® SecureCode™ program requirements associated with issuer purchase authentication and enrollment pages, and details cardholder enrollment options.

Purchase Authentication Page.....	41
Activation During Shopping Enrollment.....	41

Purchase Authentication Page

This section describes the function and requirements of the Purchase Authentication page.



General Program Requirements

A purchase authentication page is presented to authenticate cardholders when completing an online purchase at a participating Mastercard® SecureCode™ merchant.

The following requirements must be implemented on all purchase authentication pages by Mastercard SecureCode merchants.

1. The payer authentication page must be a minimum of 400 pixels high x 400 pixels wide.
 - Mastercard requires that all merchants implement the payer authentication page as an inline window. Pop-up windows are not permitted for new merchant implementations.
2. The minimum acceptable pixel size, both for the SecureCode Program Identifier and the issuer logo, is 62w x 34h. It is recommended that the issuer logo be at least as large as the SecureCode Program Identifier. The issuer logo may be larger than the SecureCode Program Identifier, however the total area designated for use by both logos must never exceed one-sixth of the total page area.

Activation During Shopping Enrollment

Activation During Shopping (ADS) is a proactive approach to cardholder enrollment when static password authentication is selected. When using stronger authentication methods such as one-time passwords (OTPs), activation during shopping is not used. In addition, risk-based authentication methods limit cardholder interruptions and reduce cardholder abandonments.

Unlike standard enrollment processes, which require that the cardholder take initiative to enroll in the program, ADS permits the issuer to solicit cardholder enrollment as part of the shopping experience. This approach also provides issuers with a means for requiring cardholder participation.

Cardholder response rates to ADS have been higher than those associated with standard enrollment, when issuers have sufficiently communicated Mastercard® *SecureCode*™ program details and benefits to their cardholder base. While effective in driving program enrollment, issuers must take care to minimize any disruption to purchase completion and avoid negative impact to participating merchants.

In recognition of this, Mastercard has outlined the requirements in this document with three goals in mind:

- **Maximize cardholder registrations**. ADS has proven to be very effective when deployed correctly. Several issuers have attained very high rates of registration with ADS.
- **Minimize loss of volume**. If not orchestrated correctly, ADS could cause some cardholder dissatisfaction. This may result in loss of purchase volume for the issuer.
- **Minimize lost merchant sales**. If merchants perceive that consumers are spending excessive time during the checkout process, they are likely to have concerns about the Mastercard *SecureCode* program.

Early use of ADS caused some concern within the merchant community regarding the potential for disruption of the sales process in some issuer deployments. Such concerns included the consumer abnormally terminating the registration, thereby causing abandonment of the entire purchase transaction.

Problems that occurred can be traced to a handful of causes that can be easily avoided:

- **Lack of consumer education**
- **Lack of proper financial institution and program identifiers**. Registration windows must clearly identify the consumer's financial institution to avoid confusion and fear of spoofing or "phishing" scams.
- **Overly long registration process**. Once presented, the registration should be short and straightforward. The use of too many questions to establish the identity of cardholders will only tax their patience and cause them to abandon the process.

Each issuer is responsible for evaluating the pros and cons of ADS enrollment in general, as well as the individual methods for initiating ADS, to determine which course of action is most appropriate for its program.

However, the requirements and guidelines outlined below are based on issuer experiences as well as on extensive focus groups and usability studies, so issuers may want to leverage these learnings for their own Mastercard *SecureCode* deployments.

Chapter 5 SecureCodeAcquirers

This section provides an overview of the activities and requirements for building and maintaining the acquirer components that are required to support Mastercard® SecureCode™.

Acquirer Overview.....	44
Acquirer Infrastructure.....	44
Establishment of SecureCode Operating Environment.....	44
Digital Signature Key Management.....	45
Authorization System Message Enhancements.....	45
Maestro Considerations.....	45
Acquirer Customization.....	47
Authentication Window Contents.....	47
Mastercard SecureCode Program Identifier Usage Guidelines.....	47
Acquirer Operational.....	48
Mastercard SecureCode Directory Server Maintenance.....	48
SSL and Digital Signing Certificates.....	48
Mastercard SecureCode 360 Webinar Series.....	48
Mastercard SecureCode Merchant Information.....	49
Mastercard SecureCode Merchant Process and Liability Shift Matrix.....	49

Acquirer Overview

The activities and requirements for acquirers are separated into the following categories.

- Infrastructure—Installation of new hardware and software components. Internal or outsourced provision of system support.
- Customization—Customizing or configuring vendor products.
- Operational—Operating the components in a production environment, including any process oriented changes that may be required.
- Accountholder Authentication Value (AAV) Processing—Handling and processing of the AAV.
- SecureCode 360—A Web presentation series available to merchants and acquirers with information on implementation and benefits of the use of Mastercard SecureCode.
- *Mastercard SecureCode—Merchant Implementation Guide*—Detailed information specific to merchants.

Acquirer Infrastructure

Following are the requirements related to the installation of new hardware and software components. If the acquirer has contracted with a Merchant Plug-In (MPI) provider, then the MPI provider will manage this.

Establishment of SecureCode Operating Environment

Following is information about establishing the Mastercard® SecureCode™ operating environment for acquirers.

Acquirers who choose to support Mastercard SecureCode on their own systems by deploying an Merchant Plug-In (MPI) must ensure compliance with the latest Mastercard specifications for authorization and clearing for all Mastercard and Maestro® products.

Acquirers can opt to not directly support the MPI and not have any infrastructure impact directly related to Mastercard SecureCode. However, if an acquirer chooses to deploy an MPI, then the acquirer must source, deploy, and test this key infrastructure element. Acquirers that want to support an MPI should refer to the *Mastercard SecureCode—Merchant Implementation Guide*. For specific questions, contact Mastercard by sending an email message to securecode_customer_support@mastercard.com

Secure Operating Facility

Acquirers may establish a secure operational facility for deployment of a server-based MPI for merchant use in order to offer an integrated solution to their merchants. All facilities must adhere to Mastercard security requirements regarding operations of cardholder authentication processing platforms.

Digital Signature Key Management

Acquirers must securely create and manage the keys to be used for performing digital signing of Merchant Plug-In (MPI) requests.

MPI hosting providers and processors must have the ability to utilize a separate digital signature key and associated Mastercard hierarchy certificate for each merchant.

MPI hosting providers and processors must have the ability to use a separate secret key for each individual merchant.

Authorization System Message Enhancements

Acquirers must adhere to Universal Cardholder Authentication Field™ (UCAF) compliance requirements as outlined in this document.

Among other things, this includes support for the UCAF and Security Level Indicator in DE 48, subelement 42 and subelement 43. Acquirers will need to ensure that they have completed validation of their authorization systems for both UCAF and Account Authentication Value (AAV) through Mastercard, and should contact Customer Implementation Services to confirm this certification has occurred.

Maestro Considerations

The following requirements and activities are specific to support of Maestro® cards as part of the Mastercard® SecureCode™ program.

Mandatory Requirements for Maestro Acceptance

Acquirers must ensure that any merchant that wants to accept Maestro must support Mastercard SecureCode on every transaction.

Account Number Length Requirements

Mastercard requires that issuers provide Maestro cardholders who want to participate in e-commerce an account number that is 13–19 digits in length and merchants that choose to accept Maestro must be able to support this variable PAN length.

Static Account Number Requirements

To facilitate both higher levels of customer service and provide increased security, Maestro requires that all Internet transactions be performed using a static account number. This number may differ from the actual account number but remains the same for all of the cardholder's Internet purchases.

Static account numbers assist merchants with their fraud management and marketing requirements when the PAN is linked to a shopper's activity. Merchants using static account numbers must ensure that they (the merchants) remain PCI-DSS compliant.

Track 2 Requirements

If the issuer's authorization system requires track 2 data, the issuer will need to implement a platform that can intercept the message prior to authorization and create real track 2 data.

Technology vendors, which implement surrogate account number solutions, are good candidates for these platforms. Currently, Maestro authorization networks will add pseudo Track 2 data in the following format.

No	Field Name	Length	Value
1	PAN	n...19	PAN as supplied in authorization message. Currently, issuers are required to issue 16 digit account numbers for Maestro Internet use. However, acquirers are requested to implement this functionality based on an account number of 12–19 digits in length.
2	Separator	ans-1	D
3	Expiry Date	ans-4	Expiry Date (YYMM) as supplied in authorization message.
4	Service Code	ans-3	101

For more information, see the appropriate authorization specification manual.

Authorization Platform Support

Maestro is supported on the Single Message System in regions outside of the Europe region. Issuers are required to reference the appropriate authorization specification manual for messaging details.

Single Message System Support

Maestro is supported on the Single Message System in regions outside of the Europe region. Acquirers are required to reference the appropriate authorization specification manual for messaging details.

Account in Good Standing

Maestro requires that all issuers provide support for Account in Good Standing transactions. For more information, refer to the [Maestro Considerations](#) section.

Acquirer Customization

The following is Mastercard® SecureCode™ customization information for acquirers.

Authentication Window Contents

The 3-D Secure protocol is designed so that creation of the cardholder authentication window is performed by the merchant.

The issuer controls the actual content of the window. All cardholder authentication pages must adhere to published Mastercard requirements on this topic. For more information, see the *Mastercard SecureCode—Issuer Implementation Guide*. Proof of adherence to the Mastercard requirements is a condition of successful completion of Mastercard® SecureCode™ issuer functional testing.

Acquirers should be aware that while they are not responsible for the content of the iFrame deployed by the issuer, Mastercard has a rigorous testing process in place to ensure a successful cardholder experience.

Mastercard SecureCode Program Identifier Usage Guidelines

Issuers and acquirers must adhere to the applicable usage guidelines.

Acquirers should advise their merchants that the merchant must adhere to the applicable usage guidelines, as well.

Merchants are required to adhere to the applicable usage guidelines.

These guidelines are indicated in the *Mastercard SecureCode Program Identifier Guidelines* accessed using the following web address:

www.mastercard.com/us/merchant/security/what_can_do/pdf/SecureCode_logo_usage.pdf

Proof of adherence to these guidelines must be provided to Mastercard as a condition of successful completion of Mastercard® SecureCode™ merchant or service provider functional testing.

Proof of adherence to these guidelines must be provided to Mastercard as a condition of successful completion of Mastercard SecureCode issuer functional testing.

Mastercard highly recommends that all screenshots be provided for review as soon as possible in case changes are required.

A copy of the Mastercard SecureCode logo artwork is available for download at <https://brand.mastercard.com/brandcenter/other-marks.html>.

A copy of the Mastercard SecureCode logo artwork, as well as any updates to the program identifier usage guidelines, is available for download at www.mastercardbrandcenter.com/us/getourbrand/index.shtml.

Acquirer Operational

Following are details about the operational aspects of acquirers implementing Mastercard® *SecureCode*™.

Mastercard *SecureCode* Directory Server Maintenance

The following responsibilities apply for issuers and acquirers for directory server maintenance.

Issuers are responsible for ensuring that Mastercard has the appropriate card range information and Access Control Server (ACS) web address for initialization of the Mastercard Directory server. When an issuer releases bank identification number (BIN) ownership, it is their responsibility to ensure the BIN is no longer enrolled with the Directory Server.

Acquirers are responsible for loading merchant IDs to the Mastercard® *SecureCode*™ Global Directory with the correct acquirer bank identification number (BIN) information. When acquirers release a BIN ownership, it is their responsibility to ensure the BIN is no longer enrolled with the Directory Server and merchant IDs have updated new acquirer BIN information.

Mastercard will accept updates only from an authorized contact, as indicated on the program enrollment forms. All updates must be made using the approved submission form, available from securecode_customer_support@mastercard.com

For more information about directory updates and maintenance schedules, refer to the *Mastercard SecureCode—Production Procedures* document.

SSL and Digital Signing Certificates

Acquirers are responsible for requesting, installing or supplying to merchants or service providers, and maintaining all necessary Secure Socket Layer (SSL) server and digital signing certificates for use by the Access Control Server (ACS) platform. Certificates are needed for communicating with the Directory Server and separately for communicating with the Mastercard Authentication History Server.

All Mastercard hierarchy certificates must be obtained from Mastercard. See the *Mastercard SecureCode—Production Procedures* manual for more information.

Mastercard *SecureCode* 360 Webinar Series

Mastercard provides a series of web-based presentations that are presented by Mastercard staff and external companies that cover a range of Mastercard® *SecureCode*™ topics.

These web presentations are provided free of charge and only require an email address to register and view the content.

Visit <https://globalrisk.mastercard.com/securecode-360/> to see available presentations and to access the service.

Mastercard SecureCode Merchant Information

Mastercard has a merchant site containing detailed information on Mastercard® SecureCode™.

Merchants can access information related to Mastercard SecureCode (for example, implementation guides) via the following URL:

- <http://www.mastercard.us/merchants/securecode.html>

Mastercard SecureCode Merchant Process and Liability Shift Matrix

The following table illustrates merchant behavior during various potential scenarios associated with Mastercard® SecureCode™ authentication request processing.

Mastercard SecureCode Processing Matrix

To access the Mastercard SecureCode Processing Matrix in a Microsoft® Excel® format that can be copied and used as needed, [click here](#). This file can be saved to a local drive for later use.

CIS DE 48 (Additional Data—Private Use) Components

These indicators, as listed in the table above, map to Mastercard Authorization specification requirements for acquirers and issuers. DE 48, subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator) contains the Security Level Indicator and DE 48, subelement 43 (Universal Cardholder Authentication Field [UCAF]) contains the UCAF Data (AAV). Merchants must refer to the appropriate acquirer or payment processor message specifications for specific formatting requirements.

Merchant Risk-Based Approach

A merchant is not required to undertake Mastercard SecureCode on every transaction except when accepting Maestro®, as this is a requirement of the program. Any merchant that chooses not to use Mastercard SecureCode on any transaction should process on the basis of the last entry in the table above labeled “Merchant Not SecureCode-enabled OR SecureCode Not Used for Transaction” (SLI = 210).

Failed Authentication

A merchant may proceed with a transaction for which the authentication failed, but it must be as a SecureCode not used for transaction (SLI = 210). This transaction must not be submitted as merchant only and is not eligible for any liability shift. No AAV will be provided, and issuers are likely to decline the transaction if submitted.

Liability Shift

For more information, refer to the *Chargeback Guide*.

Errors

Error responses are sent in lieu of the VERes. Error responses will have similar messages as iReq Reason code for the following activities.

If an error is received in response to a VReq, then the transaction must be submitted as non-Mastercard Identity Check/SecureCode and will not carry liability shift.

- Directoy Server Cache Request (CRReq) - Error 2 - message element not defined message - Mastercard does not support CRReq message
- Acquirer BIN / Merchant ID not enrolled - Error 50 - Acquirer not participating or Error 51 - Merchant not participating
- Invalid PAN - not a Mastercard valid credit / debit / prepaid card - Error 2 - Not a Mastercard supported card range
- If an error is found with the submission of the VReq due to formatting or syntax, then the authentication must be resubmitted or the transaction proceed as non-Mastercard Identity Check/SecureCode

Notes

The following numbered items correspond with the numbers provided in the “Notes” column in the table above.

1. Best practice would suggest that fraud may be involved and the merchant should prompt the consumer to try again or to use a different form of payment. If merchant decides to send for authorization, the transaction is not eligible for the liability shift and should be submitted as non-Mastercard Identity Check/SecureCode
2. The AAV associated with an attempt must be included in any subsequent authorization message.
3. The merchant should check the reason for the error message before deciding whether to proceed with the authorization. Not all reason codes may indicate a failure.
4. The PAN failed a Mod-10 check or the Mastercard Attempts service is unavailable.
5. Merchant Risk Based Approach - A merchant is not required to undertake Mastercard Identity Check/SecureCode on every transaction except when accepting Maestro, as a requirement for debit acceptance. Any merchant who chooses not to use Mastercard Identity Check/SecureCode on any transaction should process on the basis of the last entry in the matrix above entitled Merchant Does not support Mastercard Identity Check/SecureCode not used for transaction (SLI = 210).
6. The Directory Server converts the status on the ACS Verify Enrollment Responses (VERes) for N (card not enrolled) and U (unsupported device) to a Y and includes the URL for the Mastercard attempts processing Server.

Chapter 6 Issuer Authentication Options

This section introduces and describes the various authentication options available to issuers. Acquirers and merchants may be interested to know that the standard option of a static password, although present, is now being replaced by many different dynamic options improving cardholder experience.

Issuer Authentication Options Background.....	52
Static Password.....	52
Random Static.....	53
Short Message Service (SMS).....	54
Mobile Application.....	55
Chip and Pin.....	56
Display Card.....	56
Other Authentication Options.....	57

Issuer Authentication Options Background

Perhaps the most important issuer decision in a Mastercard® *SecureCode*™ implementation is how to authenticate the cardholder and, in case of a transaction dispute, provide evidence of authentication.

The decision will differ depending on the issuer's view and the attitude to risk. While issuers must evaluate the risk/reward of these options, this section will focus on explaining some of the available options.

Another consideration is whether a single authentication option fits with the issuer's strategy. The needs of an issuer's cardholders differ depending on the market segmentation of the issuer's portfolio.

Access Control Server (ACS) providers and issuers should consider the impact to cardholders and the security of authentication because weaker authentication solutions may result in fraud within fully authenticated transactions. These fraudulent transactions then become the issuer's liability upon authentication and authorization. Best practices are to use strong authentication of dynamic password programs (for example, one-time password [OTP]) with risk-based decisions to challenge cardholders appropriately while minimizing continual authentication challenges for low-risk transactions.

Additionally, the Mastercard Authentication History Server requires additional Mastercard extensions to communicate the Purchase Authentication Approach and Challenge Type for each transaction.

Static Password

Mastercard launched the Mastercard® *SecureCode*™ static password in 2002 as an entry-level initial deployment that could be quick to market, not as a long-term strategy. Much of this publication is still aimed at this proposition and, therefore, does not provide in-depth detail.

- Quick to market.
- Relies on a static password and usually on static data to identify the customer when they enroll.
- Know your customer, Identification and Verification (ID&V), is important to ensure that fraudsters can neither enroll in the service on behalf of a genuine cardholder, nor change the password of an enrolled cardholder.
- Activation During Shopping (ADS) produces good enrollment results for an issuer, but interrupts the shopping experience. It is important to follow the practices identified in this publication, because if executed poorly, ADS can cause genuine cardholders to abandon transactions. Additional information is available in the *Mastercard SecureCode Issuer Best Practices*.
- Needs communication to cardholders to ensure that they are aware of the Mastercard *SecureCode* process before they are requested to enroll, especially in the case of ADS.

Although static password was a common solution in the past, new deployments where Mastercard SecureCode deployment is strong such as the Asia/Pacific, South Asia/ Middle East/ Africa and Europe regions, are moving to a dynamic solution.

To avoid the issues surrounding ID&V some issuers using static password have undertaken a “PIN Mailing” whereby all existing cardholders receive a SecureCode through the mail with full information on its use. Making SecureCode part of the card-opening procedures is also gaining in popularity.

Random Static

Random static is a variant of a static password solution that some issuers have chosen to deploy.

It is similar to that seen for some banks Internet banking solutions. Although the cardholder still has a static password, they are never asked to complete the entire password when prompted but rather a random selection of characters from the password. For example, first, sixth, and ninth are depicted in the following figure.

MEMBER BANK

MasterCard SecureCode

Merchant: dabs.com
Amount: £35.10
Date: 18/09/00
Card Number: XXXX.XXXX.XXXX.XXXX
Personal Greeting: Vice President
Login: PAULSBAXER

Enter the first, fifth and seventh characters of your SecureCode:

[Forgot your SecureCode?](#)

This information is not shared with the merchant.

This method is an improvement in security, but makes the cardholder experience more challenging, especially if the issuer chooses to enforce password requirements such as minimum 9, or must be alphanumeric.

The same issues stated above for Static Password are also seen and should be considered in this deployment.

Short Message Service (SMS)

The use of Short Message Service (SMS) has grown in popularity with the advantage that almost any phone issued today can support SMS text messaging.

Banks use this method to send alerts to their customers, and it can now be used as an authentication method for Mastercard® *SecureCode*™.

The generation of a dynamic one-time password (OTP) sent to the registered mobile device can be undertaken in a number of ways. Issuers should discuss with their Access Control Server (ACS) service provider. One way that Mastercard undertakes this is through the use of a CAP (Chip and Pin) authentication server. This single server could be used for validation of CAP, Mobile Application and Display card values, and the generation of SMS passwords. In addition to re-using investments made in EMV, this method allows Chip issuers to examine a solution that could support several authentication options.

The advantage to the cardholder is not having to remember a password because when the card is used at an e-commerce merchant that supports Mastercard *SecureCode*, the dynamic code can be sent to the registered mobile device. Issuers could also consider alerts from their systems to the phone which advise the cardholder of the amount spent at any e-commerce merchant—even if Mastercard *SecureCode* is not used.

The cardholder experience will be similar to that of a static password as depicted in the following figure.

MEMBER BANK **MasterCard SecureCode**

Check For A New Text Message That Includes A Password For This Purchase

Merchant:	ATP Merchant
Amount:	10.00 USD
Date:	02/14/2011
Card number:	XXXX XXXX XXXX 0014
Mobile Number:	XXX-XXX-8415
Enter One Time Password from Text Message Here:	

[Hot How](#) [Resend message to Mobile](#)

[Help](#)

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It is important to ensure that the mobile device used is trusted as owned by the cardholder, and that a back-up mechanism is in place for instances when a cardholder is unable to obtain a mobile signal.

Mobile Application

Smart phones are growing in issuance and in popularity. The number of mobile applications is also growing rapidly. Mastercard has a growing number of mobile applications, some of which—such as wallets—are not yet available globally.

One option that is available globally is the Advanced Authentication for Mobile application. This application resides on the smart phone and allows a secure software application to have the equivalent of the consumers chip card on the phone. This solution allows the cardholder to have a similar experience that they would in a face-to-face chip transaction. The customer, opens the application, enters their PIN, and obtains a generated, dynamic, one-time password (OTP) to use as the *SecureCode*, which is then used in the usual box required by their issuer for authenticating the cardholder.



The issuer can verify the password by using the same “black box” solution available for any EMV-generated authentication.

Ensuring that the application is delivered to the cardholders mobile phone will be a key consideration in this deployment.

Chip and Pin

Perhaps the closest user experience to the face-to-face channel that a Chip and Pin (CAP) cardholder can experience is through the use of a CAP device as depicted in the following figure.



In this case, a cardholder would insert their card into the reader and follow the prompts on the device to enter their PIN and generate their dynamic one-time password (OTP) to use as their *SecureCode*.

Security can be increased by entering other information before entering the PIN, for example, a dynamic challenge that the issuer could display when prompting for the *SecureCode*.

Balance needs to be maintained between the user experience and the chance for fraud. For example, an issuer may decide that a simple PIN driven *SecureCode* would be sufficient for amounts under a defined threshold with increased requirements for high-value items.

This may not be a practical solution for cardholders that shop from multiple devices and locations. For those that shop either from home or work, this added security and simple cardholder experience may be appealing.

Display Card

Mastercard has launched a new card design that integrates the CAP reader into the card in two designs.

In the first design, the act of pushing a single button will generate a one time code in the display. For increased security, there is a second option that works like the CAP reader in that it requires the cardholder to first input their PIN.



These cards can be configured to also work with the CAP reader to provide both options for generating the *SecureCode*.

The Display Card is still based on the EMV standard and the same “black box” solution that would check the authentication from Mobile Authentication and CAP can be used to verify the number generated by the display card.

These cards are available for all Mastercard cards including credit, debit, and Maestro®.

Other Authentication Options

Mastercard® *SecureCode*™ has the flexibility to accommodate many authentication options that ensure both the integrity of the program and of the authentication.

Examples of other such authentication options include, but are not limited to, internet banking solutions with key fobs, secure tokens, and software tokens.

Mastercard can investigate deployment availability for issuers that have an existing authentication method. Non-standard deployments will require a thorough review by Mastercard, and agreement must be reached on screen design, and other features and options. Therefore, Mastercard should be involved early in any deployment discussion.

Send an email message for more information about additional options or an authentication concept to securecode_customer_support@mastercard.com

Virtual Card Numbers

In reality, the problem faced by an issuer is that they are liable for transactions that occur at a merchant that has deployed Mastercard *SecureCode*.

Virtual Card Numbers can be a strong issuer and cardholder solution if the numbers are generated in a secure way.

Solutions, such as the inControl offering from Mastercard, can add further value through features such as cardholder alerts, prohibitions from using the card at certain merchants/MCCs, and maximum value of purchase. This can be used for any card but is particularly popular for commercial cards.

Unfortunately a merchant can not distinguish if it is a real or a virtual card, or the level of security that may have occurred. An issuer can have an ACS that is configured to work with the inControl platform to provide the merchant with a valid fully authenticated transaction avoiding any potential acceptance issues.

When the virtual card is used, this integration avoids the cardholder being prompted for an additional *SecureCode*, and also delivers the fully authenticated response to the merchant in silent mode as a transparent authentication. The added value is that this integration can also, optionally, provide a failed authentication response to the merchant if the real card is used, and an advice to the cardholder to use the virtual card system supplied to fulfill the transaction.

For more information about inControl and *SecureCode* options, contact your Mastercard representative.

In summary, virtual cards have the following characteristics:

- Strong issuer response securing transactions for all Internet purchases both current and future. Benefits include:
 - Real card number may or may not be used for Internet purchases
 - Flexibility to be static, or a one-time generated virtual number
- Strong when integrated with Mastercard *SecureCode* by the issuer using inControl and *SecureCode* Issuer ACS for “silent” authentication.
- Weak from a merchant perspective:
 - Merchant cannot identify if the card is virtual
 - No chance of full authentication, which could trip fraud merchant parameters

Appendix A Mastercard SecureCode Contact Information

This section contains names, areas of responsibility, and contact information for Mastercard personnel who can assist customers with e-commerce enrollment, testing, and implementation issues.

Mastercard SecureCode Contact Information.....	60
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Mastercard SecureCode Contact Information

This section contains contact information for Mastercard personnel who can assist customers with e-commerce enrollment, testing, and implementation issues.

Area of Responsibility	Contact Information
Completed and signed enrollment forms	Send all completed and signed enrollment forms to: securecode_customer_support@mastercard.com
Maestro®	Customer Operations Services U.S., Canada, Caribbean, Latin America, and Middle East/Africa regions Phone: 800-999-0363 (Inside U.S.) 636-722-6176 (Outside U.S.) 636-722-6292 (Spanish Language Support) Fax: 636-722-7192 Email:securecode_customer_support@mastercard.com Europe region Phone: +32-2-352-54-03 Email:css@mastercard.com Customer Implementation Services Asia/Pacific region Email:cis_ap_support@mastercard.com Middle East/Africa region Email:cis_mea_support@mastercard.com Europe region Email:cis_europe_support@mastercard.com Canada and U.S. regions Email:cis_northamerica_support@mastercard.com Latin America and the Caribbean region Email:cis_lac_support@mastercard.com

Area of Responsibility	Contact Information
Program Management Support	securecode_customer_support@mastercard.com
Pricing/Billing	
Support	securecode_customer_support@mastercard.com

Mastercard SecureCode Online Resources

For additional information about Mastercard® SecureCode™, visit one of the following websites:

- [Mastercard Security Resources](#)
- [Mastercard SecureCode 360 Webinars](#)
- [Consumer Website](#)
- [Mastercard SecureCode Vendor List](#)
- [Mastercard SecureCode Frequently Asked Questions](#)
- [Merchant Website](#)
 - [Merchant Frequently Asked Questions](#)
 - [Program Identifier Guidelines](#)

Additional references are available on Mastercard Connect™ in English, Spanish, and Portuguese under **Library > Publications > SecureCode**.

Appendix B Maestro Processing Considerations

This section contains detailed information about specific processing issues associated with Maestro® and Mastercard® SecureCode™. Merchants should contact their acquirer for specific authorization and clearing requirements.

Account in Good Standing.....	63
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Account in Good Standing

An account in good standing transaction is a request by a merchant to check the authenticity of a Maestro® account number.

Unlike other Maestro transactions, Account in Good Standing is not a financial transaction. It does not perform a value check or guarantee that the customer has sufficient funds to cover the purchase. The objective is to satisfy the merchant's primary concern to ensure that the Maestro card number being provided by the customer is not counterfeit.

Merchants must not confuse an Account in Good Standing transaction with a pre-authorization transaction used for self-service pumps at petrol/gas stations. These transactions are used to guarantee a block of funds up to the amount in the transaction, provided it is followed within 20 minutes by a completion transaction.

An account in good standing transaction is a standard authorization message with the following specific data content requirements.

Data Element	Value
DE 4 (Amount, Transaction)	One unit of purchase currency
DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)	4 = Preauthorized Request

These data elements must be used by the acquirer when placing an Account in Good Standing transactions. Each acquirer is responsible for determining how this transaction is supported in the point of interaction message defined for the merchant to acquirer interface.

For Credit, there is the Account Status Inquiry, which is a non-financial request that allows acquirers to validate aspects of the cardholder account. Some ACS providers will also use this message to verify cardholder enrollment such as CVC, AVC, and expiry date. Additional details can be found in the *Customer Interface Specification* manual.

Data Element	Value
DE 3 (Processing Code)	00 = Purchase
DE 4 (Amount, Transaction)	Must be zero
DE 48 (Additional Data—Private Use), subelement 82 (Address Verification Service Request)	52 = AVS and authorization Request 0100
DE 48 (Additional Data—Private Use), subelement 92 (CVC 2)	CVC2 from signature panel if applicable
DE 61 (Point-of-Service [POS] Data), subfield 7 (POS Transaction Status)	8 = Account Status Inquiry Service

Appendix C Mastercard Advance Registration Program Requirements

This section defines the Mastercard Advance Registration Program (MARP) and identifies the program requirements.

Maestro Advance Registration Program.....	65
MARP Merchant Use of Mastercard <i>SecureCode</i>	65
Issuer Participation in MARP.....	66

Maestro Advance Registration Program

Global Clearing Management System supports e-commerce transactions processed under the Maestro® Advance Registration Program (MARP). MARP was introduced in 2008.

NOTE: MARP will be decommissioned 1 June 2015. For more details, refer to *Europe Region Operations Bulletin No. 2, 3 February 2014*.

MARP Merchant Use of Mastercard SecureCode

The Mastercard Advance Registration Program (MARP) enables enrolled merchants to authenticate e-commerce transactions.

NOTE: No new merchants are allowed to enroll in MARP. Effective 30 April 2014, Mastercard will be withdrawing MARP. For more details, refer to the *Europe Region Operations Bulletin No. 2, 3 February 2014*.

- The merchant invites the customer to register on its website by choosing a username and password, or similar credentials acceptable to Mastercard, and must provide the customer with the option to register a Mastercard® or Maestro® card account number for use in future e-commerce transactions.
- For the first Mastercard or Maestro e-commerce transaction, the merchant must request Mastercard® SecureCode™ authentication before submitting the transaction for authorization. If that transaction is subsequently authorized by the issuer, it is guaranteed to the acquirer or its merchant, regardless of whether the issuer or cardholder participates in Mastercard SecureCode as determined by the merchant request.
- If the first Mastercard or Maestro e-commerce transaction for the cardholder registered with the merchant is authorized by the issuer, the merchant can skip the Mastercard SecureCode authentication on subsequent transactions by the same customer using the same Mastercard or Maestro card account. In that case, the merchant will populate:
 - Data Element (DE) 48, (Additional Data Private Use, subelement 32 (Mastercard Assigned ID) in the Authorization Request/0100 message with an ID assigned by Mastercard.
 - DE 48, subelement 43 (3-D Secure for Mastercard SecureCode) in the Authorization Request/0100 message with a Mastercard-assigned static Account Authentication Value (AAV).
 - DE 48, subelement 42, position 3 (UCAF Collection Indicator) in the Authorization Request/0100 message with a value of 3.
 - Private data subelement 0052 (Electronic Commerce Security Level Indicator) subfield 3 (UCAF Collection Indicator) in the clearing record submitted to GCMS for processing (where applicable) with a value of 3.
 - The PDS 176 (Mastercard Assigned ID) in the clearing record submitted to GCMS for processing with an ID assigned by Mastercard.

- If the merchant populates the Universal Cardholder Authentication Field™ (UCAF) with the static AAV assigned by Mastercard, and populates the UCAF Collection Indicator with the value of 3, and the issuer authorizes the transaction, the issuer will have a right to charge back the transaction for reason of fraud.
- If a registered cardholder uses a different Mastercard or Maestro card account number for a transaction, the merchant must request Mastercard *SecureCode* authentication before submitting the transaction for authorization.
- Based on a risk assessment, the merchant always has the option of requesting Mastercard *SecureCode* authentication for any Mastercard or Maestro transaction, in which case the transaction will be governed by existing Mastercard *SecureCode* and Chargeback rules. For instance, for consumer cards acquired in the Europe region, if the transaction is subsequently authorized by the issuer, it is guaranteed to the acquirer or its merchant, regardless of whether the issuer or cardholder participates in Mastercard *SecureCode* as determined by the merchant request.

Issuer Participation in MARP

Any Maestro® issuer that supports e-commerce is required to support the static Accountholder Authentication Value (AAV) in authorization until 1 June 2015, at which time Mastercard will discontinue the Maestro Advance Registration Program (MARP). For more details, refer to *Europe Region Operations Bulletin* No. 2, 3 February 2014.

Issuers must meet existing Europe region requirements to support Maestro e-commerce transactions.

Requirements can be found in the *Authorization Manual* and, if applicable, the *GCMS Reference Manual* available on the Publications page on Mastercard Connect™.

Appendix D India IVR Transactions (SecureTelephone)

This section provides an overview of the Mastercard requirements to support IVR Transactions in India.

Overview.....	68
Data Extensions to the Existing 3-D Secure Protocol.....	68
UCAF Transport in Mastercard Authorization Messages.....	68
Mastercard <i>SecureCode</i> —Security Level Indicator (DE 48, subelement 42).....	69
Universal Cardholder Authentication Field (DE 48, subelement 43).....	70
What is an AAV?.....	70
Sample IVR Transaction Flow.....	71
Mastercard <i>SecureCode</i> Compliance and Functional Testing.....	71

Overview

Following the successful deployment of 3-D Secure (SecureCode) for all domestic electronic commerce transactions, the banking authority of India, Reserve Bank of India (RBI), has defined a mandate that requires a similar two-factor authentication process to also be rolled out for IVR transactions.

As there was no technology or precedent of authenticating an IVR transaction with two-factor authentication, Mastercard has worked with IVR vendors to utilize existing investments in technology, process and infrastructure to build a framework and specification using the 3-D Secure protocol. As owner of the 3-D Secure Protocol, Visa has published a country-specific (India) specification that defines a number of additional data elements in the existing 3-D Secure messages.

For more information, refer to the Visa document *3-D Secure Functional Requirements—Extensions for Mobile and IVR Transactions* in India v1.1.

Data Extensions to the Existing 3-D Secure Protocol

As detailed in the 3-D Secure Functional Requirements—Extensions for Mobile and IVR Transactions in India v1.1, the Verify Enrollment Request (VEReq), Verify Enrollment Response (VERes), and PAREq messaging within the 3-D Secure protocol have been extended to allow the Merchant plug-in (MPI) and Issuer Access Control Server (ACS) to convey the additional transaction related elements that identify an IVR transaction, as opposed to an electronic commerce transaction.

UCAF Transport in Mastercard Authorization Messages

Mastercard has designated specific subelements within DE 61 (Point-of-Service [POS] Data) and DE 48 (Additional Data—Private Use) for the identification of SecureTelephone Order and transport of Universal Cardholder Authentication Field™ (UCAF)-related data and associated indicators in authorization messages.

These subelements will be used to support and identify IVR transactions within the authorization message. Support for SecureTelephone Order within the authorization message was mandated as part of the 7.2 Banknet release.

The subelements are described in the following sections. Refer to the *Customer Interface Specification* manual for additional information regarding authorization message formatting.

SecureTelephone—DE 61—Point-of-Service (POS) Data, subelements 4, 7, and 10. The following subelement values are to correctly identify an IVR (SecureTelephone Order) DE 61.

Position	Value	Description
4—POS Cardholder Presence	3	Cardholder Not Present, Phone/ARU Order
7—POS Transaction Status	2	SecureCode Phone Order
10—Cardholder-Activated Terminal Level	MUST NOT BE 6	Authorized Level 6 CAT:Electronic Commerce

Mastercard *SecureCode*—Security Level Indicator (DE 48, subelement 42)

The SecureCode Security Level Indicator contains the fields representing the security protocol and cardholder authentication type associated with the transaction.

Mastercard requires that DE 48 (Additional Data—Private Use), subelement 42 (Electronic Commerce Indicators) be included and properly populated for all electronic commerce and SecureTelephone (IVR) transaction authorizations.

Please note that only Fully authenticated IVR transactions (Security Level Indicator 2 [UCAF data collection is supported by the merchant, and Universal Cardholder Authentication Field™ {UCAF} data is present in DE 48, subelement 43 [Universal Cardholder Authentication Field {UCAF}]] are applicable to SecureTelephone order (IVR) transactions.

The required DE 48 (Additional Data—Private Use), subelement 42 (Electronic Commerce Indicators), subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator) values for SecureTelephone order (IVR) are provided in the table below.

Position	Value	Description
1—Security Protocol	2	Channel
2—Cardholder Authentication	1	Cardholder certificate not used
3—UCAF Collection Indicator	0	UCAF data collection is not supported by the merchant
	1	UCAF data collection is supported by the merchant, and UCAF data may be available (DE 48, subelement 43 may be present for Mastercard <i>SecureCode</i>)
	2	UCAF data collection is supported by the merchant, and UCAF data must be present (DE 48, subelement 43)

Universal Cardholder Authentication Field (DE 48, subelement 43)

The Universal Cardholder Authentication Field™ (UCAF) contains the encoded Mastercard® *SecureCode*™ issuer-generated authentication data (collected by the merchant) resulting from all Mastercard *SecureCode* fully authenticated transactions.

UCAF is a standard, globally interoperable method of collecting cardholder authentication data at the point of interaction. Within the Mastercard authorization networks, UCAF is a universal, multipurpose data transport infrastructure that is used to communicate authentication information between cardholders, merchants, issuers, and acquirers. It is a variable length (maximum 32 characters) field with a flexible data structure that can be tailored to support the needs of issuer security and authentication approaches.

NOTE: All acquirers and issuers must ensure that they can send and/or receive contents in this field up to the maximum length of 32. Note that applications utilizing this field, such as Mastercard *SecureCode*, may provide contents that are less than the maximum length. For Mastercard *SecureCode* specifically, this field will be 28 positions. This field should NOT include any padding to meet the maximum length of 32 bytes.

What is an AAV?

The Accountholder Authentication Value (AAV) is a Mastercard® *SecureCode*™ specific token that uses the Universal Cardholder Authentication Field™ (UCAF) field for transport within Mastercard authorization messages.

It is generated by the issuer and presented to the merchant for placement in the authorization request. This AAV can be proof of a fully authenticated or an attempted authentication transaction.

The AAV is used to identify the processing parameters associated with the transaction. Among other things, the field values will identify the:

- Issuer ACS that created the AAV. (This could be the Issuer ACS or, in the case of an attempt, the Mastercard Attempt processing server.)
- Sequence number that can positively identify the transaction for that location
- Secret key used to create the Message Authentication Code (MAC), which is a cryptographic method that ensures AAV data integrity, and binds the entire AAV structure to a specific PAN.

UCAF is the mechanism that is used to transmit the AAV from the merchant to issuer for authentication purposes during the authorization process.

MARP Exception

For merchants that have qualified to participate in the Mastercard Advance Registration Program (MARP), Mastercard assigns a static AAV that conforms to the same format as an issuer-generated AAV.

NOTE: MARP is closed to new business and will be decommissioned 1 June 2015. For more details, refer to *Europe Region Operations Bulletin No. 2*, 3 February 2014.

Sample IVR Transaction Flow

An example of a SecureTelephone order (IVR) transaction is as follows.

1. Cardholder calls the merchant to order items, and finalize purchase.
2. Merchant collects all necessary data, including the cardholder's primary account number (PAN) and phone information.
3. The merchant's modified IVR-MPI creates a Verify Enrollment Request (VEReq) message with the IVR extensions, including the following data:
 - a. Format of Phone number or Device ID
 - b. Cardholder Phone number or Device ID
 - c. PAREq channel—DIRECT
 - d. Shopping Channel—IVR
 - e. Available Authentication Channels
4. The Mastercard Directory Server will forward the VEReq message to the appropriate Issuer IVR-ACS, to validate that the PAN is enrolled in the service.
5. The issuer IVR-ACS responds to the Mastercard Directory with confirmation of enrollment and the Verify Enrollment Response (VERes) message including the ACS web address is returned to the Merchant IVR-MPI.
6. IVR-MPI generates a PAREq message with the IVR extension, and sends to the appropriate Issuer IVR-ACS.
7. ACS receives and processes the PAREq message—IVR extensions may be used by the Issuer ACS in the authentication process.
8. Upon successful validation of the cardholder (or using data contained in the extended PAREq), the issuer ACS will generate the PAREs message and forward to Merchant IVR-MPI.
9. IVR-MPI receives PAREs and proceeds with authorization request.

Mastercard *SecureCode* Compliance and Functional Testing

Mastercard has developed vendor compliance testing, as well as issuer and merchant functional testing, to ensure vendor products and member and merchant implementations are compliant and successfully interoperate with all Mastercard® *SecureCode*™ platforms and infrastructure.

Proof of Visa Certification of IVR-ACS and IVR-MPI development is required before Mastercard compliance testing can commence.

All vendors, merchant endpoints, and issuers are required to complete the designated testing process prior to launch. More information regarding testing is available from securecode_customer_support@mastercard.com

Appendix E Mastercard Extensions for the Brazil Market

This section provides specifications related to extensions that support Brazil domestic processing. These extensions allow passing of data from the merchant to the issuer in 3-D Secure messages.

Brazil Market Extensions.....	73
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Brazil Market Extensions

This section provides specifications related to extensions that support Brazil domestic processing. These extensions allow passing of data from the merchant to the issuer in 3-D Secure messages.

These extensions will be used in both the VEReq and PAReq messages.

- Card Account Type
- Mobile Phone Number
- Merchant Category Code
- Transaction Type

The VEReq and PAReq are extended to allow the Merchant Plug In (MPI) to pass the additional transaction-related elements to the Access Control Server (ACS) specified below.

Field Definitions for Brazil Market Extensions in VEReq and PAReq

Element Name	Description	Format and Values
Extension Id	visa.3ds.brazil_market	
brazilmcc	Merchant Category Code, provided by the merchant	Length 1–4, Numeric digits
brazilaccounttype	Account Type as selected by the cardholder during checkout.	Length 1–2, Numeric digits • 00 (NOT APPLICABLE) • 01 (CREDIT) • 02 (DEBIT)
brazilmobilenum	Mobile number (as entered by the cardholder during checkout)	Length 1–25, Numeric digits
braziltransactiontype	Transaction Type, provided by the merchant	Length 1–2; Numeric Digit • 00 (Goods/Service Purchase) • 03 (Check Acceptance) • 10 (Account Funding) • 11 (Quasi-Cash Transaction) • 28 (Prepaid Activation & Load)

NOTE: The critical attribute must be set to “false” for all Extension fields defined in this protocol extension specification.

Message Sample of Brazil Market Extensions in VReq

```
<Extension id="visa.3ds.brazil_market" critical="false">  
  <brazilmcc>Brazil MCC</brazilmcc>  
  <brazilaccounttype>Brazil Card Account Type</brazilaccounttype>  
  <brazilmobilenum>Brazil Cardholder Mobile Number</brazilmobilenum>  
  <braziltransactiontype>Brazil Transaction Type</braziltransactiontype>  
</Extension>
```

Message Sample of Brazil Market Extensions in PReq

```
<Extension id="visa.3ds.brazil_market" critical="false">  
  <brazilmcc>Brazil MCC</brazilmcc>  
  <brazilaccounttype>Brazil Card Account Type</brazilaccounttype>  
  <brazilmobilenum>Brazil Cardholder Mobile Number</brazilmobilenum>  
  <braziltransactiontype>Brazil Transaction Type</braziltransactiontype>  
</Extension>
```

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